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Implementation of a Transformer-Based Architecture Using Cochleograms for Abnormal Respiratory Sound Classification

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Abstract

Respiratory diseases represent a major global health burden, and their early detection through pulmonary auscultation remains highly dependent on clinical expertise. This project investigates automated abnormal respiratory sound classification using transformer-based architectures combined with cochleogram representations, evaluated on the ICBHI 2017 dataset containing 6898 respiratory cycles from 126 patients.

The cochleogram, computed using a gammatone filterbank with 64 filters, provides a biologically-inspired time-frequency representation that captures respiratory sound characteristics more accurately than conventional spectrograms. The resulting representations are converted to 3-channel RGB images using the Viridis colormap and used as input to the classification models.

Several architectures are investigated and compared, starting with convolutional neural networks as baselines, followed by a Vision Transformer (ViT) trained from scratch. To address the significant class imbalance inherent to the dataset, optimizations not explored in the original work are introduced, including focal loss, weighted random sampling, and class-balanced training strategies. Additionally, the Audio Spectrogram Transformer (AST) is explored as an alternative transformer architecture for this task.

All models are evaluated using a 10-fold cross-validation strategy grouped by patient to prevent data leakage. The proposed optimizations yield improved sensitivity in detecting adventitious sounds, demonstrating the potential of transformer-based architectures combined with cochleogram representations for reliable automated respiratory sound classification.

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1 General Context

The lungs are among the most vital and vulnerable organs of the human body. Every breath carries not only oxygen but also pathogens, pollutants, and allergens capable of triggering disease. Despite remarkable advances in medical imaging and laboratory diagnostics, the simple act of listening to the lungs through a stethoscope remains one of the most immediate and informative clinical assessments available to a physician. Yet this act is fragile—dependent on experience, attention, and conditions that vary from one consultation to the next. This project is born from a straightforward question: *can a machine learn to hear what a doctor hears, and do so consistently, at scale, and without fatigue?*

1.1 The Silent Epidemic: Why Respiratory Diseases Matter

Respiratory diseases are among the leading causes of death and disability worldwide, yet they receive comparatively little public attention relative to their true burden. According to the Global Burden of Disease Study 2023, chronic respiratory diseases collectively affected an estimated **569.2 million people** and caused **4.2 million deaths** in 2023 alone [1]. Lower respiratory infections—chiefly pneumonia—remain the world’s **leading infectious cause of death**, killing millions annually, with children under five and elderly populations bearing the heaviest toll [2].

What makes this burden particularly tragic is that many of these deaths are preventable. Early detection is the cornerstone of effective respiratory disease management: a patient diagnosed with pneumonia in its early stages has dramatically better outcomes than one who presents in respiratory failure. The same is true for COPD, asthma, and pulmonary fibrosis—conditions that progress silently for years before manifesting as clinical emergencies. The challenge, therefore, is not only medical but also logistical: how do we bring reliable, early diagnostic capability to every patient, everywhere?

1.1.1 Chronic Obstructive Pulmonary Disease

Chronic Obstructive Pulmonary Disease (COPD) is a slowly progressive, largely irreversible condition caused by sustained airway inflammation and structural remodeling, most commonly triggered by tobacco smoke and particulate air pollution. It is the **third leading cause of death worldwide**, responsible for approximately **3.7 million deaths** in 2021, with an age-standardized mortality rate of 45.2 per 100,000 [3]. More than **400 million people** live with COPD globally, with prevalence reaching 10.6% among adults aged 40 and older [4].

Despite a 37% decline in age-standardized death rates between 1990 and 2021, the absolute number of COPD deaths has *increased* by 49% over the same period—driven by population growth and aging [5]. The burden falls disproportionately on low- and middle-income countries, where over 80% of deaths occur. Particulate matter pollution has recently surpassed smoking as the leading global risk factor, contributing 41.7% of COPD-related DALYs [3]. Acoustically, COPD exacerbations are

characterized by *wheezes*—continuous, high-pitched sounds produced by turbulent airflow through narrowed, obstructed airways.

1.1.2 Asthma

Asthma is the most prevalent chronic respiratory disease in the world, affecting an estimated **363 million people** in 2023 and causing approximately **442,000 deaths** annually [6]. Unlike COPD, asthma is characterized by reversible airway obstruction and disproportionately affects younger populations—global prevalence among children and adolescents reached **130.52 million cases** in 2023 [7]. Between one-third and one-half of patients experience severe symptoms that regularly interfere with daily life, including sleep, school attendance, and physical activity [8].

The hallmark auscultatory finding in asthma is the expiratory *whheeze*, produced by the oscillation of airway walls during forced expiration against a partially obstructed lumen. In severe exacerbations, wheezes may be audible throughout both inspiration and expiration, reflecting the degree of bronchospasm. The automated detection of wheezes carries direct clinical implications: in telemedicine and remote monitoring settings, it could enable the early identification of deteriorating asthma control before a crisis occurs.

1.1.3 Pneumonia

Pneumonia is a life-threatening acute infection of the lung parenchyma caused by bacteria, viruses, or fungi. *Streptococcus pneumoniae* alone accounts for 25.3% of all lower respiratory infection deaths globally [2]. The disease kills nearly **three million children** annually—more than HIV and malaria combined in the under-five age group [9]. In adults, particularly the elderly and immunocompromised, pneumonia remains a leading cause of intensive care unit admission and mortality.

The acoustic signature of pneumonia is the *crackle*—a discontinuous, explosive sound produced by the sudden opening of fluid-filled or collapsed airways during inspiration. Fine, late-inspiratory crackles are particularly characteristic of early pneumonia, while coarser crackles suggest more extensive consolidation or bronchiectasis [10]. The ability to detect crackles objectively and reliably has direct triage implications, especially in emergency and primary care settings where access to chest radiography may be limited.

1.1.4 Pulmonary Fibrosis

Idiopathic pulmonary fibrosis (IPF) is a chronic, progressive, and ultimately fatal interstitial lung disease in which normal lung tissue is replaced by fibrotic scar tissue, leading to inexorable decline in respiratory function. Its global incidence ranges from 3 to 9 cases per 100,000 per year in Europe and North America, with crude mortality rates rising from 2.10 per 100,000 in 2001 to 3.14 per 100,000 by 2022 [11]. The median survival following diagnosis is only 3 to 5 years, and no curative treatment currently exists.

IPF produces characteristic fine, late-inspiratory crackles often described as “Velcro-like”—a sound produced by the sudden opening of fibrotic, stiffened airways. These crackles can precede radiological evidence of fibrosis by months or years, making auscultation-based detection a potentially powerful early screening tool [12].

1.2 The Stethoscope and Its Limits

Two centuries after René Laënnec rolled a sheet of paper into the first stethoscope in 1816, auscultation remains a cornerstone of clinical practice—non-invasive, instantaneous, and extraordinarily informative when performed by a skilled examiner. A population-based study found that over a **quarter of adults** present adventitious lung sounds on auscultation, with crackles predominantly heard during inspiration and wheezes during expiration [13]. These findings carry meaningful clinical information: the presence of adventitious sounds at multiple auscultation sites is associated with significantly higher risk of reduced lung function, even in apparently healthy individuals.

Yet auscultation’s greatest strength—its immediacy—is also its greatest weakness. The interpretation of respiratory sounds is inherently subjective. Studies comparing clinician performance on identical audio recordings have documented alarming variability: overall auscultation accuracy ranges from **60.3% for medical students** to **80.1% for senior fellows**, and inter-observer agreement on crackle identification is substantially lower than for wheezes [14]. Environmental noise, patient body habitus, stethoscope quality, and clinician fatigue all introduce additional sources of error.

In resource-limited settings—which bear the greatest burden of respiratory disease—these limitations are compounded by the scarcity of specialist physicians. A patient in a rural clinic may be evaluated by a general practitioner or community health worker with limited auscultation training, using a basic stethoscope, in a noisy environment. The consequences of diagnostic errors in this context are severe and largely preventable.

This diagnostic gap is the central motivation for the development of automated respiratory sound classification systems—tools capable of objectively analyzing auscultatory recordings and providing consistent, reproducible diagnostic support regardless of the examiner’s expertise or the clinical setting.

1.3 From Signal to Diagnosis: The Role of Artificial Intelligence

The advent of digital stethoscopes has transformed respiratory sound analysis from an ephemeral clinical act into a reproducible, storable, and computationally tractable signal processing problem. A digital recording captures the full acoustic content of a respiratory cycle—its frequency content, temporal envelope, and fine-grained transient structure—enabling the application of machine learning algorithms to extract patterns invisible to the human ear.

Early automated approaches relied on hand-crafted features such as Mel-Frequency Cepstral Coefficients (MFCCs), wavelet coefficients, and linear predictive coding, combined with classical classi-

fiers such as Support Vector Machines and Hidden Markov Models. These methods yielded modest performance and required substantial domain expertise in acoustic signal processing.

The emergence of deep learning fundamentally transformed the landscape. Convolutional Neural Networks (CNNs), adapted from computer vision, demonstrated the ability to learn discriminative acoustic features directly from time-frequency representations of audio signals—effectively treating spectrograms as images. More recently, Transformer-based architectures [15]—originally developed for natural language processing—have been adapted for both visual and acoustic classification tasks. The Vision Transformer (ViT) [16] and the Audio Spectrogram Transformer (AST) [17] represent the state of the art in this evolution, capturing long-range temporal dependencies in acoustic signals through multi-head self-attention mechanisms that CNNs cannot model natively.

A critical and often overlooked dimension of this problem is the choice of **acoustic representation**. The quality of the input fed to a classifier determines the ceiling of its performance. Conventional spectrograms impose a fixed time-frequency resolution that may fail to capture the fine temporal structure of crackles or the sustained oscillatory character of wheezes. The **cochleogram**—a biologically-inspired representation computed using a gammatone filterbank that mimics the frequency resolution of the human cochlea—offers a compelling alternative. By distributing frequency channels according to the Equivalent Rectangular Bandwidth (ERB) scale, the cochleogram provides non-uniform spectral resolution that is naturally suited to the acoustic characteristics of adventitious respiratory sounds.

1.4 The ICBHI 2017 Challenge

The **ICBHI 2017 Scientific Challenge**, organized by the International Conference on Biomedical and Health Informatics, was a landmark event in the field of respiratory sound analysis. Its primary objective was to establish a standardized benchmark for the automated classification of lung sounds, providing the research community with a common dataset, evaluation protocol, and baseline against which proposed systems could be fairly compared.

Prior to ICBHI 2017, research in respiratory sound classification was fragmented — studies used heterogeneous private datasets, inconsistent annotation schemes, and incompatible evaluation metrics, making cross-study comparison virtually impossible. The challenge addressed this gap by releasing a large, publicly available, and carefully annotated collection of respiratory recordings, along with a clearly defined scoring metric based on the average of sensitivity and specificity.

Since its release, the ICBHI 2017 dataset has become the de facto standard for evaluating respiratory sound classification algorithms, accumulating a substantial body of published work spanning traditional machine learning approaches, convolutional neural networks, and more recently transformer-based architectures. Reported scores on this benchmark range widely depending on the methodology employed, reflecting both the genuine difficulty of the task and the lack of consensus on evaluation practices across studies.

This work uses the ICBHI 2017 dataset as its experimental foundation. A detailed description of the dataset’s composition, class distribution, and inherent challenges is provided in Chapter 2.

1.5 Objectives

This project investigates whether the combination of a biologically-grounded acoustic representation—the cochleogram—with modern transformer-based deep learning architectures can yield a reliable, clinically meaningful classifier for respiratory sound analysis on the ICBHI 2017 benchmark. The work takes as its primary reference the study by [18], which proposed the CochleogramViT architecture, and aims to reproduce, improve upon, and critically compare against its reported results. Specifically, the work pursues four concrete objectives:

1. **Establish CNN baselines.** Implement and evaluate a baseline CNN and an AlexNet-based classifier on cochleogram representations to provide additional reference performance points consistent with the existing literature, and to assess the limitations of convolutional approaches relative to transformer-based models.
2. **Investigate transformer architectures.** Train and evaluate a Vision Transformer (ViT) from scratch and an Audio Spectrogram Transformer (AST) on cochleogram inputs, assessing whether the self-attention mechanism provides meaningful advantages over convolutional inductive biases for this task.
3. **Address class imbalance through targeted optimizations.** Introduce and systematically evaluate a combination of training strategies — including weighted cross-entropy loss, softened class weight scaling, class-balanced batch sampling, and learning rate scheduling — designed to improve sensitivity for minority classes without sacrificing specificity, with direct attention to the clinical implications of false negatives.
4. **Improve the sensitivity-specificity trade-off.** Systematically investigate the tension between sensitivity and specificity that arises from aggressive imbalance correction, and identify training configurations that achieve a clinically acceptable balance between detecting true adventitious sounds and avoiding false alarms on healthy patients.
5. **Ensure rigorous, leakage-free evaluation.** Apply a patient-independent 10-fold cross-validation protocol that guarantees complete separation of patients across training and validation sets, producing results that genuinely reflect generalization to unseen patients rather than memorization of patient-specific acoustic signatures.

1.6 Report Organization

The remainder of this report is structured as follows. The first section surveys the state of the art in automated respiratory sound classification, covering signal representations, deep learning architectures, and key results on the ICBHI 2017 benchmark. The second section details the methodology: cochleogram generation, model architectures, training strategies, and evaluation protocol. The third section presents and analyzes the experimental results across all evaluated configurations. The final section concludes the report with a synthesis of findings and directions for future research.

2 Problem Statement

Automating the classification of respiratory sounds is a task that sits at the intersection of clinical necessity and technical difficulty. While the motivation for such a system is clear — reducing dependence on subjective auscultation and extending diagnostic reach — the path from raw lung sound recordings to a reliable classifier is obstructed by a set of interrelated challenges that must be explicitly acknowledged and addressed. This chapter defines the core problems that the present work confronts.

2.1 Difficulty of Respiratory Sound Classification

Respiratory sound classification is inherently more difficult than many standard audio classification tasks for several reasons. First, the acoustic differences between sound categories are subtle and often context-dependent. Crackles are characterized by brief, discontinuous bursts of energy, while wheezes are continuous, musical tones superimposed on the breathing cycle — but in practice these characteristics overlap, vary in intensity, and may be partially masked by environmental noise or patient movement artifacts.

Second, respiratory sounds are recorded under highly variable conditions. The ICBHI 2017 dataset, which serves as the experimental foundation of this work, contains recordings from seven different stethoscope models across multiple auscultation sites and clinical settings. This device and environment variability introduces acoustic inconsistencies that are unrelated to the underlying pathology, forcing any classifier to learn representations that are robust to recording conditions rather than specific to them.

Third, inter-patient variability is substantial. The acoustic signature of a wheeze in a child differs from that of an elderly patient with COPD, and both may differ from the same sound recorded at a different thoracic location. A model trained on one patient group must generalize to unseen patients with potentially different acoustic profiles — a requirement that places strict demands on the evaluation protocol and the representational capacity of the model.

2.2 Class Imbalance

The ICBHI 2017 dataset exhibits a pronounced class imbalance that constitutes one of the most consequential technical challenges in this work. As shown in Table 1 of Chapter 1, Normal respiratory cycles account for 52.8% of all annotated samples, while the rarest class — Both, indicating the simultaneous presence of crackles and wheezes — represents only 7.3%. Crackles and Wheezes occupy intermediate positions at 27.0% and 12.8% respectively.

This distribution has a direct and measurable effect on model behavior. A classifier that predicts Normal for every input would achieve approximately 52.8% accuracy while contributing nothing diagnostically useful. More critically, such a model would exhibit near-zero sensitivity — the ability to detect genuinely abnormal sounds — which is the metric of primary clinical importance in a screening context.

Standard training procedures without any imbalance correction consistently produce this majority-class collapse behavior. The model minimizes the cross-entropy loss most efficiently by concentrating its predictive capacity on the dominant class, leaving minority classes — particularly Wheezes and Both — effectively unlearned. This is not a failure of the architecture but a consequence of the optimization objective operating on skewed data.

Addressing class imbalance is therefore not an optional refinement but a prerequisite for producing a clinically meaningful classifier. However, overcorrection carries its own risk: excessively aggressive reweighting pushes the model in the opposite direction, predicting every sample as adventitious and collapsing specificity to near zero. The practical challenge is finding a correction strategy that lifts minority class sensitivity without destroying the model’s ability to correctly identify healthy patients — a balance that proves delicate in practice and is one of the central empirical investigations of this work.

2.3 Need for Robust Audio Representations

The choice of audio representation has a fundamental impact on what a deep learning model can and cannot learn. Raw waveforms contain the complete acoustic signal but present the model with an extremely high-dimensional, temporally fine-grained input that is difficult to learn from with limited data. Standard spectrograms provide a time-frequency decomposition but treat all frequency bands with equal resolution, which does not reflect the non-linear frequency sensitivity of the human auditory system.

Mel spectrograms partially address this by warping the frequency axis according to the mel scale, emphasizing lower frequencies where much perceptually relevant information is concentrated. However, the mel scale is derived from psychoacoustic studies of speech and general hearing, not from the specific frequency characteristics of respiratory sounds, which occupy a relatively narrow band between approximately 100 Hz and 2000 Hz.

Cochleograms offer an alternative grounded in auditory physiology. Generated using a gammatone filterbank that mimics the frequency selectivity of the human cochlea, cochleograms provide a biologically motivated time-frequency representation that may be better suited to capturing the subtle spectro-temporal patterns that distinguish normal from adventitious respiratory sounds. The reference work upon which this project is based [18] proposed cochleograms as the input representation for a Vision Transformer, hypothesizing that the biological plausibility of the representation would contribute to classification performance.

A key question this work inherits from the reference paper is whether cochleograms genuinely provide an advantage over alternative representations for this specific task, or whether the choice of representation is secondary to the training strategy and architecture. This question motivates the careful experimental design employed throughout and informs the interpretation of results presented in later chapters.

3 Chapter 1 — Background and State of the Art

3.1 Respiratory Sound Analysis

Respiratory sound analysis is the study of acoustic signals produced by airflow through the lungs and airways during the breathing cycle. These sounds, captured through auscultation using a stethoscope placed at various locations on the thorax, carry clinically significant information about the state of the respiratory system. Abnormalities in airflow, caused by narrowing, obstruction, or fluid accumulation in the airways, manifest as characteristic acoustic patterns that trained clinicians learn to identify and associate with specific pathological conditions.

The systematic study of respiratory sounds dates back to the early 19th century with the invention of the stethoscope by René Laennec, and the terminology used to describe lung sounds has been refined over nearly two centuries of clinical practice. Despite this long history, the interpretation of respiratory sounds remains highly subjective, with significant inter-observer variability even among experienced pulmonologists. This subjectivity, combined with the growing availability of digital recording equipment and machine learning tools, has motivated a substantial body of research into automated respiratory sound analysis over the past two decades.

3.1.1 Normal Respiratory Sounds

Normal respiratory sounds are produced by turbulent airflow in the larger airways and transmitted through the lung parenchyma to the chest wall. They are characterized by a soft, continuous, low-frequency profile during inspiration, with expiration typically quieter and shorter. Over the chest wall of healthy individuals, normal vesicular breath sounds occupy a frequency range of approximately 100 to 500 Hz, with the power spectrum decreasing exponentially as frequency increases [19]. The absence of superimposed acoustic events is the defining characteristic of a normal respiratory cycle and serves as the baseline against which adventitious sounds are identified.

3.1.2 Crackles

Crackles are discontinuous, non-musical adventitious sounds superimposed on the normal breathing pattern. They are produced by the sudden opening of previously collapsed or fluid-filled small airways and alveoli during inspiration, generating brief bursts of acoustic energy heard as popping or crackling sounds. Crackles are clinically associated with conditions such as pneumonia, pulmonary fibrosis, bronchiectasis, and congestive heart failure.

From a signal processing perspective, crackles are characterized by their impulsive, transient nature — with individual events lasting less than 20 ms and a frequency content spanning 60 to 2000 Hz [20]. Fine crackles, associated with interstitial lung disease, tend to be shorter in duration and higher in pitch, while coarse crackles, more commonly seen in COPD and bronchiectasis, are longer and lower-pitched [21].

3.1.3 Wheezes

Wheezes are continuous, musical adventitious sounds with a sinusoidal waveform and a dominant frequency of 400 Hz or higher [20]. They are produced by the oscillation of airway walls in narrowed or partially obstructed bronchi, generating a tonal quality that distinguishes them perceptually from both normal sounds and crackles. Individual wheeze events typically last between 80 and 250 ms [22], and their presence is a key diagnostic indicator of airway obstruction in conditions such as asthma and COPD.

In time-frequency representations, wheezes appear as horizontal striations — sustained energy concentrations at specific frequencies that persist across a significant portion of the breathing cycle. Despite this distinctive appearance, wheezes are among the most difficult classes to classify reliably in the ICBHI 2017 dataset, partly due to their relative scarcity and partly due to acoustic overlap with other sound types in certain patient populations.

3.1.4 Combined Abnormal Sounds

The fourth class in the ICBHI 2017 dataset, referred to as Both, denotes respiratory cycles in which crackles and wheezes are simultaneously present. This class represents the most acoustically complex category, as the classifier must recognize the co-occurrence of two distinct adventitious sound types within a single cycle. Patients presenting with both sound types typically suffer from more advanced or multi-factorial respiratory conditions, and the simultaneous presence of crackles and wheezes carries important diagnostic and prognostic significance.

From a machine learning perspective, the Both class presents the additional challenge of being the rarest in the dataset, accounting for only 7.3% of all annotated cycles. This scarcity, combined with its acoustic complexity, makes it the most difficult class to learn reliably and the one most susceptible to being ignored by models trained without explicit imbalance correction.

Table 1: Acoustic characteristics of the four ICBHI 2017 respiratory sound classes

Property	Normal	Crackles	Wheezes	Both
Temporal pattern	Continuous	Discontinuous	Continuous	Mixed
Duration	Full cycle	<20 ms	80–250 ms	Variable
Frequency range	100–2000 Hz	60–2000 Hz	100–1600 Hz	Overlapping
Dominant frequency	<500 Hz	Broadband	≥ 400 Hz	Broadband
Waveform type	Broadband noise	Impulsive bursts	Sinusoidal	Composite
Clinical association	Healthy	Pneumonia, fibrosis, bronchiectasis	Asthma, COPD	Advanced or multi-factorial

3.2 ICBHI 2017 Respiratory Sound Database

The ICBHI 2017 Respiratory Sound Database [23] is the experimental foundation of this work and the most widely used benchmark for respiratory sound classification research. It was compiled independently by two research teams across two countries over several years: the School of Health Sciences at the University of Aveiro (ESSUA), Portugal, which collected recordings at the Respiratory Research and Rehabilitation Laboratory and at Hospital Infante D. Pedro, Aveiro; and a joint team from the Aristotle University of Thessaloniki (AUTH) and the University of Coimbra (UC), which acquired recordings at the Papanikolaou General Hospital, Thessaloniki, and the General Hospital of Imathia, Greece [23].

The key statistics of the database are summarized in Table 2.

Table 2: Key statistics of the ICBHI 2017 Respiratory Sound Database

Property	Value
Total duration	5.5 hours
Total audio recordings	920
Total patients	126
Total respiratory cycles	6,898
Patient age range	Children to elderly
Recording duration range	10 – 90 seconds
Cycle duration range	0.2 – 16 seconds
Number of recording devices	4
Sampling rate range	4,000 – 44,100 Hz
Number of clinical sites	4
Number of disease categories	7 + healthy

3.2.1 Dataset Description

The database contains a total of 5.5 hours of annotated respiratory sound recordings distributed across 920 audio files from 126 subjects, spanning children to elderly individuals [23]. Each audio file corresponds to a single patient recording at a specific auscultation site, and the full dataset encompasses 6,898 annotated respiratory cycles. The respiratory cycles were manually annotated by respiratory experts and cross-reviewed to ensure label reliability [24].

Recordings were acquired using four digital stethoscope models with heterogeneous sampling rates, introducing device-level acoustic variability that constitutes one of the dataset’s primary challenges for automated classification. The recording equipment and their corresponding sampling rates are summarized in Table 3.

Recordings were collected from multiple anatomical auscultation sites on the chest wall, including

Table 3: Recording devices used in the ICBHI 2017 dataset

Device	Sampling Rate
AKG C417L Microphone	44,100 Hz
3M Littmann Classic II SE	4,000 Hz
3M Littmann 3200	4,000 Hz
Welch Allyn Meditron	10,000 Hz

anterior, posterior, and lateral locations. Recording durations range from 10 to 90 seconds, and individual respiratory cycle durations range from 0.2 to 16 seconds, reflecting the natural variability in patient breathing patterns [23]. Noise levels in some cycles are intentionally retained to simulate real clinical conditions.

The patient population covers a broad clinical spectrum. Subjects were diagnosed with one of seven respiratory pathological conditions or were healthy controls, as summarized in Table 4.

Table 4: Patient diagnoses in the ICBHI 2017 dataset

Diagnosis	Type
Chronic Obstructive Pulmonary Disease (COPD)	Chronic
Bronchiectasis	Chronic
Asthma	Chronic
Bronchiolitis	Non-chronic
Pneumonia	Non-chronic
Upper Respiratory Tract Infection (URTI)	Non-chronic
Lower Respiratory Tract Infection (LRTI)	Non-chronic
Healthy	No pathology

3.2.2 Class Distribution

The 6,898 respiratory cycles in the dataset are annotated into four sound classes based on the presence or absence of adventitious sounds. The class distribution is shown in Table 5.

The distribution reveals a pronounced imbalance that has significant implications for classifier design. Normal cycles constitute over half the dataset, while the rarest class — Both, indicating simultaneous crackles and wheezes — accounts for only 7.3% of all cycles. This four-to-one ratio between the most and least frequent classes means that a naive classifier defaulting to the majority class would achieve 52.8% accuracy while contributing nothing diagnostically useful.

The challenge is compounded by the patient-level grouping of recordings. Since multiple cycles are extracted from the same patient recording, a random train-test split would allow patient-specific

Table 5: Class distribution of the ICBHI 2017 dataset

Class	Cycles	Proportion
Normal	3,642	52.8%
Crackles	1,864	27.0%
Wheezes	886	12.8%
Both	506	7.3%
Total	6,898	100%

acoustic signatures to leak between training and evaluation sets, producing optimistically biased performance estimates. A patient-independent split — in which all recordings from a given patient appear exclusively in either the training or the evaluation set — is therefore essential for producing results that genuinely reflect generalization to unseen patients. This requirement directly motivates the grouped cross-validation protocol employed in this work and described in Chapter 4.

3.3 Audio Representations

A critical design decision in any audio classification pipeline is the choice of input representation. Raw audio waveforms, while containing the complete acoustic signal, are high-dimensional, temporally fine-grained sequences that are difficult for deep learning models to learn from directly, particularly with limited training data. Time-frequency representations, which decompose the signal into its frequency components over time, have therefore become the dominant approach for feeding audio signals into image-based deep learning architectures such as CNNs and Vision Transformers.

Several time-frequency representations have been proposed and evaluated in the respiratory sound classification literature, each reflecting different assumptions about which acoustic properties are most relevant to the task. The three most commonly used are spectrograms, mel spectrograms, and cochleograms, which differ primarily in how they distribute frequency resolution across the spectrum.

3.3.1 Spectrograms

The spectrogram is the most direct time-frequency representation, computed by applying the Short-Time Fourier Transform (STFT) to successive overlapping windows of the audio signal. Each column of the resulting two-dimensional matrix represents the power spectrum of the signal within one time window, and the full spectrogram captures how spectral energy evolves over time.

Spectrograms distribute frequency resolution linearly — equal bandwidth is allocated to each frequency bin regardless of its perceptual or clinical relevance. For respiratory sounds, which are concentrated in the 100–2000 Hz range, a linear frequency axis allocates significant representa-

tional capacity to higher frequencies that carry little diagnostic information, while potentially under-resolving the clinically relevant lower frequency bands where crackles and wheezes are most prominent.

3.3.2 Mel Spectrograms

The mel spectrogram addresses the linear frequency limitation by warping the frequency axis according to the mel scale, a perceptual scale derived from psychoacoustic studies of human hearing. The mel scale approximates the non-linear frequency resolution of the human auditory system, allocating more frequency bins to lower frequencies and fewer to higher frequencies, reflecting the greater perceptual sensitivity of the human ear at lower pitches.

Mel spectrograms have become the standard input representation for many audio deep learning tasks, including speech recognition, music classification, and environmental sound detection. Their application to respiratory sound classification is straightforward and has produced competitive results in the literature. However, the mel scale is derived from speech perception studies and is not specifically optimized for the frequency range and characteristics of respiratory sounds. The filterbank cutoff frequencies and bandwidth allocations of the mel scale may therefore not align optimally with the acoustic features that distinguish crackles from wheezes in lung auscultation recordings.

3.3.3 Cochleograms

Cochleograms are time-frequency representations grounded in auditory physiology rather than general perceptual studies. They are generated using a gammatone filterbank, a bank of bandpass filters whose center frequencies and bandwidths are derived from measurements of the mechanical response of the human cochlea. The gammatone filterbank provides a biologically motivated frequency decomposition that captures the non-linear, frequency-selective behavior of the inner ear more accurately than either the linear STFT or the mel scale approximation.

For respiratory sound analysis, the cochleogram’s filterbank is typically configured to cover the 100–2000 Hz range that encompasses the clinically relevant acoustic content of normal and adventitious respiratory sounds. The resulting representation emphasizes the frequency bands most relevant to respiratory physiology, potentially providing a more discriminative input for classifiers trained to distinguish between sound classes that differ subtly in their spectro-temporal structure.

The reference work upon which this project is based [18] proposed cochleograms as the input representation for a Vision Transformer, motivated by the hypothesis that a biologically grounded representation would better capture the perceptually meaningful differences between respiratory sound classes. A detailed description of the cochleogram generation pipeline implemented in this work is provided in Chapter 4.

Table 6 summarizes the key properties of the three representations discussed above.

Table 6: Comparison of audio representations for respiratory sound classification

Property	Spectrogram	Mel Spectrogram	Cochleogram
Frequency axis	Linear	Mel scale	Gammatone (cochlear)
Biological basis	None	Speech perception	Cochlear physiology
Frequency resolution	Uniform	Speech-optimized	Auditory-optimized
Low-freq emphasis	No	Partial	Yes
Clinically motivated	No	Partial	Yes
Common in RSC literature	Yes	Yes	Emerging
Used in this work	No	No	Yes

3.4 Deep Learning for Audio Classification

The application of deep learning to audio classification has evolved rapidly over the past decade, driven by the availability of large annotated datasets and advances in neural network architectures. For respiratory sound classification in particular, deep learning has progressively displaced traditional machine learning approaches based on hand-crafted features such as Hidden Markov Models, Support Vector Machines, and Gaussian Mixture Models, which required extensive domain expertise and struggled to generalize across heterogeneous recording conditions [18].

3.4.1 Convolutional Neural Networks

Convolutional Neural Networks (CNNs) were the first deep learning architecture to demonstrate strong performance on audio classification tasks, and they remain the dominant approach in the respiratory sound classification literature [18]. The key insight enabling this transfer from image classification to audio is the representation of audio signals as two-dimensional time-frequency images — spectrograms, mel spectrograms, or cochleograms — which CNNs can process using the same convolutional operations developed for visual recognition.

CNNs exploit local spatial correlations through learnable convolutional filters, enabling them to detect time-localized spectral patterns such as the impulsive bursts characteristic of crackles or the sustained harmonic trajectories of wheezes. Their hierarchical architecture, stacking multiple convolutional and pooling layers, allows progressively more abstract features to be extracted from the raw time-frequency representation.

Several CNN architectures have been applied to respiratory sound classification on the ICBHI 2017 dataset. The baseline CNN proposed alongside the dataset consists of two convolutional layers followed by a deep neural network layer with leaky ReLU activations [18]. AlexNet, one of the first deep CNNs to achieve breakthrough performance on large-scale image recognition, features five convolutional layers, three pooling layers, and three fully connected layers [18]. VGG16 extends this with 16 convolutional layers emphasizing small 3×3 filters, while ResNet50 introduced residual connections to enable training of very deep 50-layer networks. Despite their differences in depth

and architecture, all CNN-based approaches share the limitation of relying on local receptive fields, which may miss long-range temporal dependencies in respiratory sound patterns.

3.4.2 Vision Transformers

The Vision Transformer (ViT), introduced by Dosovitskiy et al. [25], applied the transformer architecture originally developed for natural language processing to image classification by treating an image as a sequence of fixed-size patches. Each patch is linearly embedded into a token, positional encodings are added to preserve spatial information, and the resulting sequence is processed by a standard transformer encoder using multi-head self-attention mechanisms.

The self-attention mechanism enables ViT to model global dependencies between all patches simultaneously, regardless of their spatial distance — a fundamental advantage over CNNs whose receptive field grows only gradually through successive layers. For audio classification, where relevant acoustic events may span the entire duration of a recording, this global context modelling is particularly valuable.

Applied to time-frequency representations of respiratory sounds, ViT processes cochleogram patches and learns to attend to spectro-temporal regions that are most discriminative for each sound class. The reference work [18] demonstrated that this combination of cochleogram representation and ViT architecture outperforms all evaluated CNN baselines on the ICBHI 2017 dataset, achieving an accuracy of 67.9% in the four-class classification scenario, compared to 62.9% for the best CNN baseline.

3.4.3 Audio Spectrogram Transformer

The Audio Spectrogram Transformer (AST), proposed by Gong et al. [26], extends the ViT architecture specifically for audio tasks by leveraging pretraining on large-scale audio datasets. Unlike the ViT trained from scratch, AST initializes its weights from a model pretrained on ImageNet and then fine-tunes on AudioSet — a large-scale dataset of over 2 million audio clips spanning 527 sound categories. This pretraining strategy addresses the data scarcity problem that limits the performance of transformers trained from scratch on small medical datasets such as ICBHI 2017.

AST processes log mel spectrogram patches and has demonstrated state-of-the-art performance on several audio classification benchmarks. Its application to respiratory sound classification is an active area of research, motivated by the hypothesis that the rich acoustic representations learned during AudioSet pretraining can transfer effectively to the task of detecting adventitious respiratory sounds.

3.4.4 Sharpness-Aware Minimization

Sharpness-Aware Minimization (SAM) [27] is an optimization strategy designed to improve model generalization by seeking parameters that lie in flat regions of the loss landscape rather than sharp

minima. The key insight is that models converging to sharp minima — where the loss increases steeply in all directions — tend to generalize poorly to unseen data, while flat minima — where the loss remains low in a broad neighborhood — tend to generalize better.

SAM achieves this by simultaneously minimizing the loss value and its sensitivity to parameter perturbations, requiring two forward-backward passes per training step. While this doubles the computational cost per epoch, the generalization improvements are significant, particularly for small datasets where overfitting is a persistent challenge. For a model with 13 million parameters trained on approximately 6,200 samples per fold — as in this work — the regularization effect of SAM is a compelling motivation for its investigation.

3.5 Reference Work

The primary reference for this project is the study by Mang et al. [18], published in *Sensors* in January 2024, titled *Classification of Adventitious Sounds Combining Cochleogram and Vision Transformers*. This work, conducted at the University of Jaen and the University of Oviedo, Spain, constitutes the first published application of the cochleogram–ViT combination to adventitious respiratory sound classification, and serves as both the methodological foundation and the performance benchmark against which the contributions of the present work are evaluated.

3.5.1 Review of the Reference Paper

The central contribution of Mang et al. is the systematic investigation of four time-frequency representations — STFT spectrogram, Mel-Frequency Cepstral Coefficients (MFCC), Constant-Q Transform (CQT), and cochleogram — as inputs to five deep learning architectures: BaselineCNN, AlexNet, VGG16, ResNet50, and a Vision Transformer. The study evaluates all combinations on the ICBHI 2017 dataset under both binary (adventitious vs. normal) and four-class (Normal, Crackles, Wheezes, Both) classification scenarios, using 10-fold cross-validation with patient-independent splits.

The authors’ hypothesis is that the cochleogram, by virtue of its biologically motivated gammatone filterbank, provides a more discriminative time-frequency representation for respiratory sounds than standard alternatives, and that the global self-attention mechanism of the ViT is better suited to capturing the spectro-temporal patterns of adventitious sounds than the local receptive fields of CNNs.

3.5.2 Cochleogram Generation

In the reference paper, cochleograms are generated using a gammatone filterbank of $K = 64$ filters with centre frequencies uniformly distributed between 100 Hz and $f_s/2$ Hz along the Equivalent Rectangular Bandwidth (ERB) scale. A filter order of $o = 4$ is used, consistent with standard cochlear modelling practice. Each respiratory cycle is standardized to 6 seconds via zero-padding, downsampled to 4 kHz since respiratory sounds do not exceed 2 kHz, and the resulting cochleogram

is converted to a three-channel RGB image using the Viridis colormap for compatibility with standard image-based deep learning architectures [18].

3.5.3 ViT Architecture

The ViT architecture employed in the reference paper closely follows the original ViT design [25]. The cochleogram image is partitioned into fixed-size patches, each linearly embedded into a token of dimension $d_{\text{model}} = 512$. Positional embeddings are added, and the sequence is processed by a transformer encoder consisting of $N = 6$ identical layers, each containing a multi-head self-attention sub-layer with 8 attention heads and a position-wise feed-forward network with hidden dimension 1024. A learnable classification token is prepended to the sequence, and its output representation at the final layer is passed to a softmax classification head producing probabilities over the four respiratory sound classes [18].

3.5.4 Reported Results

The results reported by Mang et al. demonstrate the superiority of the cochleogram–ViT combination over all evaluated baselines. In the four-class classification scenario on the full ICBHI 2017 dataset, the proposed method achieves the results summarized in Table 7.

Table 7: Four-class classification results reported by Mang et al. [18] on the ICBHI 2017 dataset. Sen = Sensitivity, Spe = Specificity, Sco = Score, Acc = Accuracy.

Model	Sen (%)	Spe (%)	Sco (%)	Acc (%)
AlexNet + Cochleogram	45.12	59.75	52.43	54.48
ResNet50 + Cochleogram	41.78	57.78	49.78	52.31
VGG16 + Cochleogram	53.45	68.71	61.08	62.94
BaselineCNN + Cochleogram	52.71	68.84	60.78	62.93
ViT + Cochleogram	56.77	71.37	64.03	67.99

The ViT combined with cochleogram achieves the best performance across all metrics, with a Score of 64.03% and an accuracy of 67.99%, statistically significantly outperforming all CNN baselines as confirmed by Mann–Whitney U and Wilcoxon signed-rank tests [18].

A critical observation, explicitly acknowledged by the authors, is that sensitivity values are consistently lower than specificity values across all architectures and representations. This reflects the majority-class bias induced by the class imbalance in the ICBHI 2017 dataset — the models learn to correctly identify Normal sounds more reliably than adventitious sounds. The authors note this as a limitation but do not propose explicit strategies to address it. This gap constitutes the primary motivation for the imbalance handling contributions of the present work.

4 Chapter 2 — Proposed Methodology

4.1 Overall Framework

The proposed system addresses abnormal respiratory sound classification as a three-stage pipeline: audio-to-cochleogram transformation, transformer-based classification, and patient-independent evaluation.

In the first stage, raw respiratory cycle recordings from the ICBHI 2017 dataset are converted into cochleogram representations. Each audio cycle is processed through a gammatone filterbank with 64 frequency channels, followed by sub-band envelope extraction, power-law compression, and resizing to a 128×128 image. This biologically-inspired preprocessing replaces the conventional mel-spectrogram, as the gammatone filterbank more faithfully models the frequency resolution of the human cochlea and better captures the transient structure of crackles and the sustained tonal character of wheezes.

In the second stage, the cochleogram is treated as an image input to a transformer-based classifier. Two architectures are evaluated: a Vision Transformer (ViT) trained from scratch using the cochleogram as an RGB image, and an Audio Spectrogram Transformer (AST) pretrained on AudioSet, which operates directly on the single-channel cochleogram. Both architectures divide the input into 16×16 patches and apply multi-head self-attention across patch embeddings. To address the pronounced class imbalance of the ICBHI dataset, weighted cross-entropy loss with a softened class weighting scheme is applied during training. For the best-performing AST configuration, Sharpness-Aware Minimization (SAM) is additionally employed as the outer optimizer.

In the third stage, all models are evaluated using 10-fold StratifiedGroupKFold cross-validation, which ensures complete patient-level separation between training and validation sets while maintaining class distribution balance across folds. Performance is reported using the ICBHI score—the average of sensitivity and specificity under the binary normal-versus-adventitious convention—computed both as a per-fold mean with standard deviation and as a pooled aggregate across all folds.

4.2 Cochleogram Generation Pipeline

4.2.1 Audio Preprocessing

Each respiratory cycle is extracted from its parent recording using the annotated start and end timestamps provided by the ICBHI dataset. The raw waveform is loaded via `torchaudio`, converted to mono by averaging channels if necessary, and resampled to the target sampling rate specified in the configuration, typically 22,050 Hz. The extracted clip is then padded with zeros or truncated to a fixed duration controlled by the `clip_duration` parameter. This ensures that all inputs to the cochleogram pipeline have identical length regardless of the original cycle duration, which can vary from 0.2 to 16 seconds in the ICBHI dataset.

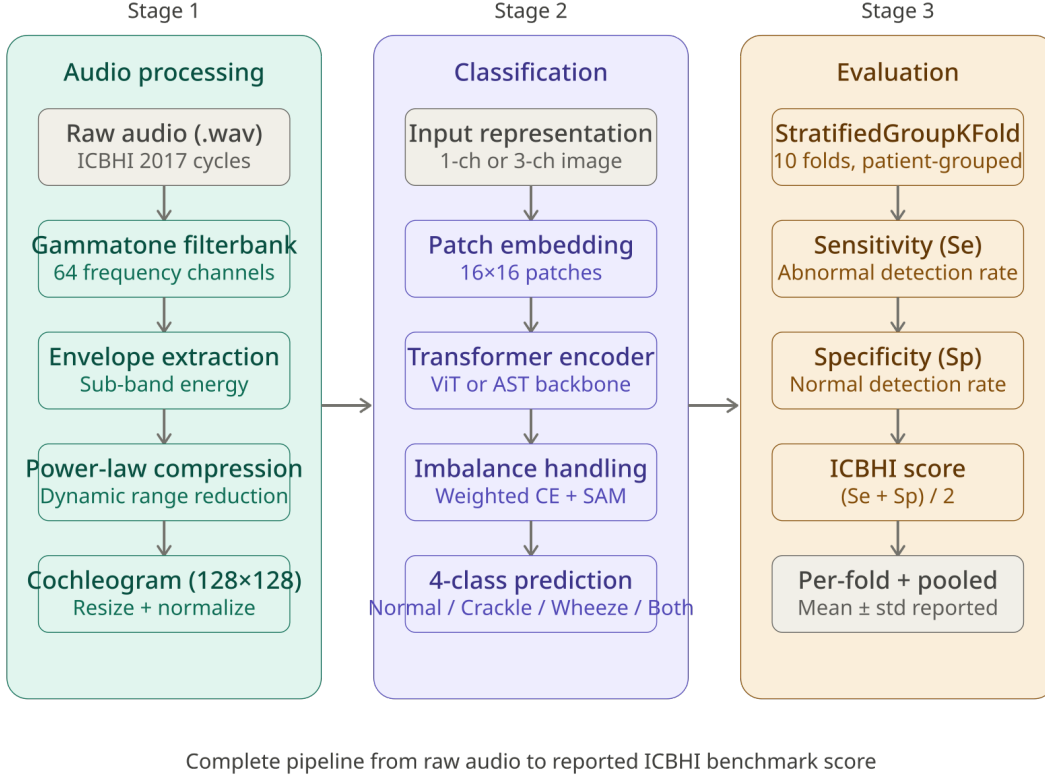


Figure 1: Complete system pipeline

4.2.2 Gammatone Filterbank

The preprocessed waveform is passed through a gammatone filterbank implemented via the `pycochleagram` library, following the McDermott Lab pipeline. The gammatone filter is designed to emulate the behavior of the human cochlea by using non-uniform spectral resolution: lower frequencies are represented with narrower bandwidths, while higher frequencies are represented with broader bandwidths. This adaptive time-frequency resolution makes the cochleogram robust to noise and acoustic variability, and is particularly suitable for respiratory sounds whose clinically relevant components are concentrated in the low-frequency range.

The filterbank decomposes the signal into K frequency channels. In this work, $K = 64$ gammatone filters are used, with center frequencies $f_c(k)$ distributed on the Equivalent Rectangular Bandwidth (ERB) scale between 100 Hz and $f_s/2$ Hz. This range covers the dominant spectral content of adventitious respiratory sounds, especially crackles and wheezes. A gammatone filter impulse response is obtained by multiplying a gamma distribution by a sinusoidal carrier:

$$g(t) = at^{o-1}e^{-2\pi b(f_c)t}\cos(2\pi f_ct + \phi), \quad t > 0, \quad (1)$$

where a is an amplitude factor, o is the filter order, f_c is the center frequency in Hertz, ϕ is the phase, and $b(f_c)$ is the exponential decay coefficient controlling the filter bandwidth. In this work, an order of $o = 4$ is used because it provides a satisfactory approximation of the human auditory

filter. The bandwidth term is defined from the ERB scale as

$$b(f_c) = 1.019 \text{ERB}(f_c), \quad (2)$$

with

$$\text{ERB}(f_c) = 24.7 \left(4.37 \frac{f_c}{1000} + 1 \right). \quad (3)$$

ERB spacing mimics the frequency selectivity of the human basilar membrane, placing more channels in the low-frequency range where respiratory acoustic patterns are most informative. Filtering is performed via FFT convolution, which is computationally efficient for the signal lengths encountered in this dataset.

4.2.3 Envelope Extraction

After filtering, the instantaneous amplitude envelope of each sub-band is extracted using the analytic signal through the Hilbert transform. This discards the carrier frequency of each channel and retains only the temporal energy modulation, the component most relevant to distinguishing transient crackle bursts from sustained wheeze oscillations. For each frequency channel and analysis frame, a cochleogram coefficient is then computed by aggregating the filtered sub-band energy over a window:

$$C(k, m) = \sum_{n=0}^{N-1} \hat{X}(k, n) w(n), \quad (4)$$

where $\hat{X}(k, n)$ denotes the gammatone-filtered signal for the k -th channel, $k = 1, \dots, K$, $w(n)$ is the analysis window, N is the window length, and $C(k, m)$ is the coefficient associated with center frequency $f_c(k)$ for the m -th frame. The resulting cochleogram is therefore a two-dimensional time-frequency representation with shape `(n_filters, n_time_frames)`.

4.2.4 Compression

A power-law non-linearity with exponent 0.3 is applied element-wise to the envelope matrix:

$$\hat{E}(f, t) = E(f, t)^{0.3}. \quad (5)$$

This compressive non-linearity serves two purposes. First, it reduces the dynamic range of the representation, preventing loud transient events from dominating the input and improving robustness across recordings with different microphone gains. Second, it approximates the compressive response of the inner hair cells of the cochlea, improving the biological fidelity of the representation. Following compression, the entire array is normalized to the range $[0, 1]$ per sample.

4.2.5 Resizing

The compressed envelope matrix is resized to a fixed spatial dimension of 128×128 pixels using bilinear interpolation, implemented through `CochleogramTransform`. This step decouples the cochleogram resolution from the original signal length and sample rate, ensuring all model inputs share a consistent shape regardless of cycle duration.

4.2.6 Normalization and Storage

The final 128×128 array is stored as a `float32` NumPy file (`.npy`) on disk. A metadata CSV is generated alongside the cochleograms, recording the source WAV path, cycle start and end times, class label, and path to the corresponding `.npy` file. This preprocessing is performed once offline; at training time, the dataset loader reads directly from the `.npy` files, avoiding redundant computation. For the AST model, the single-channel cochleogram is used as-is; for the ViT baseline, it is converted to a 3-channel RGB image by applying the Viridis colormap.

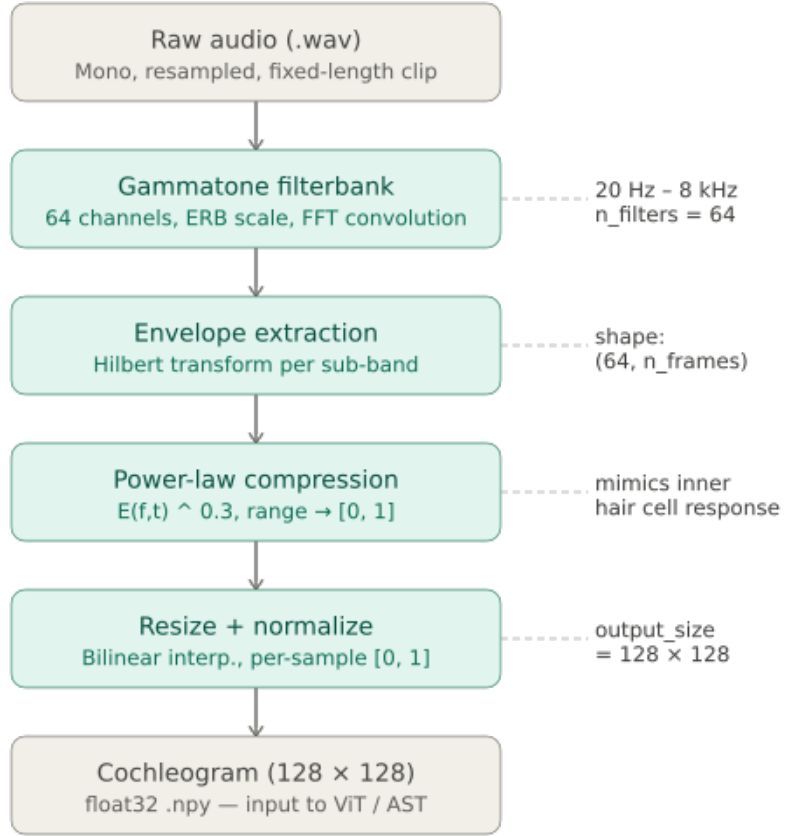


Figure 2: Cochleogram generation workflow

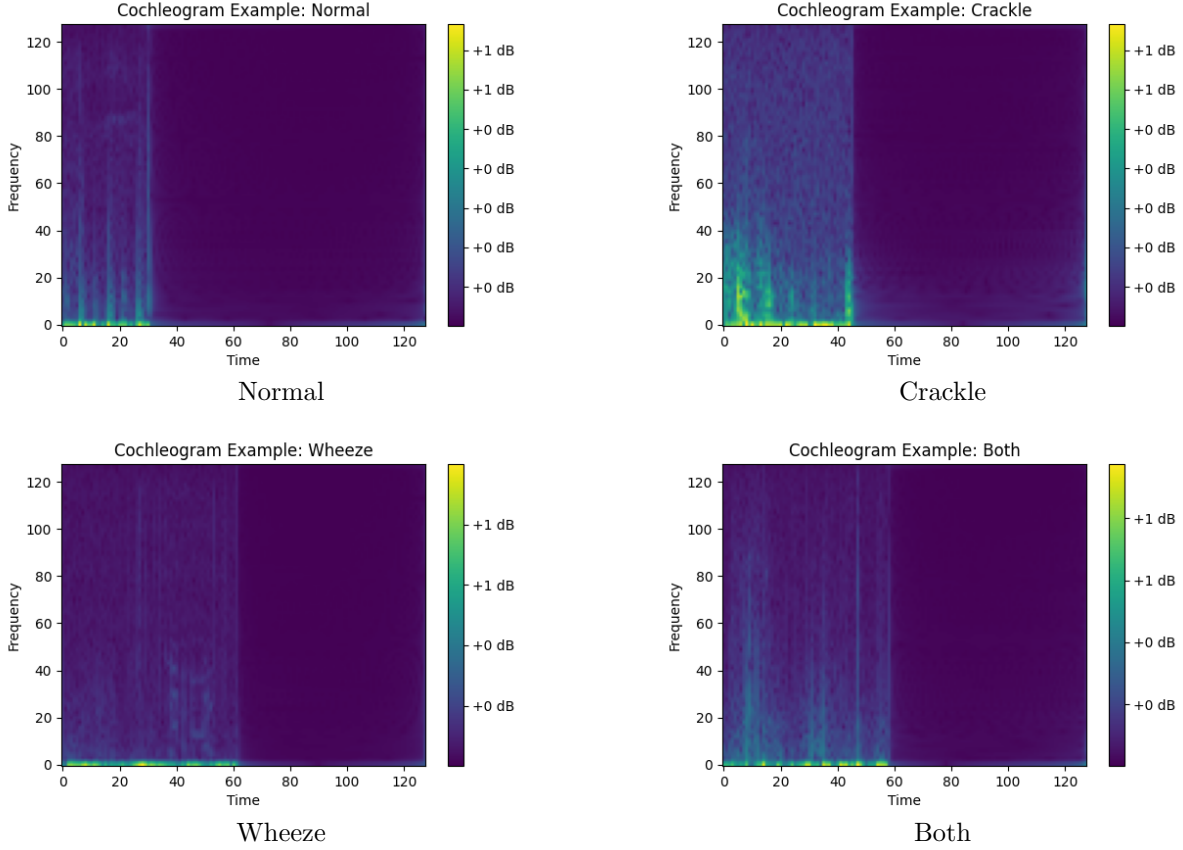


Figure 3: Example cochleogram representations for each ICBHI class in the dataset

4.3 Data Imbalance Handling

4.3.1 Dataset Imbalance Analysis

The ICBHI 2017 dataset exhibits a pronounced class imbalance that constitutes one of the primary technical challenges of this work. The Normal class accounts for 52.8% of all respiratory cycles, corresponding to 3,642 samples, while the rarest class—Both, representing simultaneous crackles and wheezes—accounts for only 7.3%, corresponding to 506 samples. This roughly 7:1 ratio between the majority and minority classes means that a naive classifier which always predicts Normal achieves high specificity at near-zero sensitivity, a trade-off that is clinically unacceptable. In a medical screening context, the cost of a false negative—a sick patient classified as healthy—far outweighs the cost of a false positive. Addressing this imbalance is therefore both a technical and a patient safety requirement.

4.3.2 Weighted Cross-Entropy Loss

The primary imbalance correction strategy adopted in this work is a weighted cross-entropy loss function. Standard cross-entropy treats all classes equally, so the gradient signal is dominated by the majority Normal class during training. Weighted cross-entropy assigns a per-class scalar weight w_c to the loss contribution of each sample, so that the model pays proportionally more attention to minority-class errors:

$$\mathcal{L} = - \sum_{c=1}^C w_c \cdot y_c \log \hat{y}_c. \quad (6)$$

The class weights are derived from the inverse class frequencies, then passed through a softening function controlled by the `soften_power` hyperparameter p :

$$w_c = \left(\frac{N}{N_c} \right)^p, \quad (7)$$

where N is the total number of samples and N_c is the count of class c . Setting $p = 1$ recovers full inverse-frequency weighting, while $p = 0$ collapses all weights to 1, corresponding to no correction. In all final experiments, $p = 0.25$ is used, which applies a gentle push toward minority classes without destabilizing training on the Normal class.

4.3.3 Weighted Random Sampling

A weighted random sampler modifies mini-batch construction by assigning each training sample a sampling probability proportional to the inverse frequency of its class. This increases the expected frequency of minority-class examples in each batch, creating an approximately balanced class distribution at the batch level rather than at the loss level.

This strategy was evaluated in combination with weighted loss, corresponding to double correction, and in isolation. Ablation experiments revealed that applying both corrections simultaneously caused specificity collapse: the model over-predicted adventitious sounds, producing a flood of false positives and driving specificity to approximately 27%, close to the 25% expected from a uniform random predictor over four classes. This failure mode arises because both mechanisms independently push the model toward a balanced prior of 25% per class, while the true data distribution is approximately 47% Normal. The combined effect over-corrects, causing the model to systematically under-predict the Normal class.

4.3.4 Softened Class Weights

The softening exponent p controls the aggressiveness of the minority-class correction. A full inverse-frequency weighting with $p = 1$ applies very large weights to the Both and Wheeze classes, which risks overfitting to the few available samples of those classes and suppressing Normal predictions. A softened exponent $p < 1$ tempers this effect. Experiments with $p \in \{0.25, 0.5, 0.75\}$ showed that $p = 0.75$ reproduced the specificity collapse observed with double correction, $p = 0.5$ recovered balanced Se/Sp but offered no aggregate score improvement over $p = 0.25$, and $p = 0.25$ produced

the most stable training dynamics across folds. The value $p = 0.25$ was therefore adopted for all reported configurations.

4.4 Vision Transformer Baseline

4.4.1 Architecture

The Vision Transformer (ViT) baseline closely follows the original transformer principle of representing an input as a sequence of vectors and processing that sequence using stacked self-attention and position-wise feed-forward layers. In the original neural sequence transduction setting, an encoder maps an input sequence of symbol representations $(x_1, \dots, x_n)$ into continuous representations $z = (z_1, \dots, z_n)$, and a decoder autoregressively generates an output sequence $(y_1, \dots, y_m)$. For image classification, ViT adapts this idea by using only the encoder-style transformer stack: the image is converted into a sequence of patch embeddings, positional information is added, and the resulting sequence is processed by transformer encoder blocks.

In this work, the cochleogram is treated as an image-like representation. For the ViT baseline, each single-channel cochleogram is converted into a 3-channel RGB image using the Viridis colormap, allowing the model to operate with the same input convention as standard vision models. The implemented model, denoted **CochleogramViT**, uses 128×128 RGB inputs, a 16×16 patch size, four output classes, an embedding dimension of 512, six transformer encoder layers, eight attention heads, an MLP hidden dimension of 1024, dropout of 0.3 inside the transformer blocks, and embedding dropout of 0.2. The resulting architecture is shown in Figure 4.

4.4.2 Patch Embedding

The cochleogram image is partitioned into non-overlapping fixed-size patches of 16×16 pixels. Each patch is flattened and projected through a trainable linear embedding layer, producing a vector representation for that patch. A learnable classification token, denoted [CLS], is prepended to the patch sequence so that its final representation can summarize the full input for classification. Learnable positional embeddings are then added to the patch embeddings to preserve spatial ordering:

$$z_0 = [x_{\text{cls}}; x_p^1 E; x_p^2 E; \dots; x_p^N E] + E_{\text{pos}}, \quad (8)$$

where x_p^i is the i -th flattened image patch, E is the linear projection matrix, N is the number of patches, x_{cls} is the classification token, and E_{pos} is the positional embedding matrix.

4.4.3 Transformer Encoder

The embedded patch sequence is passed through a stack of transformer encoder layers. Each layer contains two main sub-layers: a multi-head self-attention (MSA) block and a position-wise multilayer perceptron (MLP). Residual connections and layer normalization are applied around

each sub-layer to improve information flow and gradient propagation. The encoder update can be written as

$$z'_\ell = \text{MSA}(\text{LN}(z_{\ell-1})) + z_{\ell-1}, \quad (9)$$

$$z_\ell = \text{MLP}(\text{LN}(z'_\ell)) + z'_\ell, \quad (10)$$

where ℓ indexes the transformer layer. This structure allows the model to learn long-range dependencies between cochleogram regions, rather than relying only on local convolutional neighborhoods.

4.4.4 Classification Head

After the final transformer encoder layer, the output representation corresponding to the classification token is passed to a linear classification head. This head predicts one of the four respiratory sound classes: Normal, Crackles, Wheezes, or Both. The model is trained end-to-end using the weighted cross-entropy objective described in the imbalance handling section.

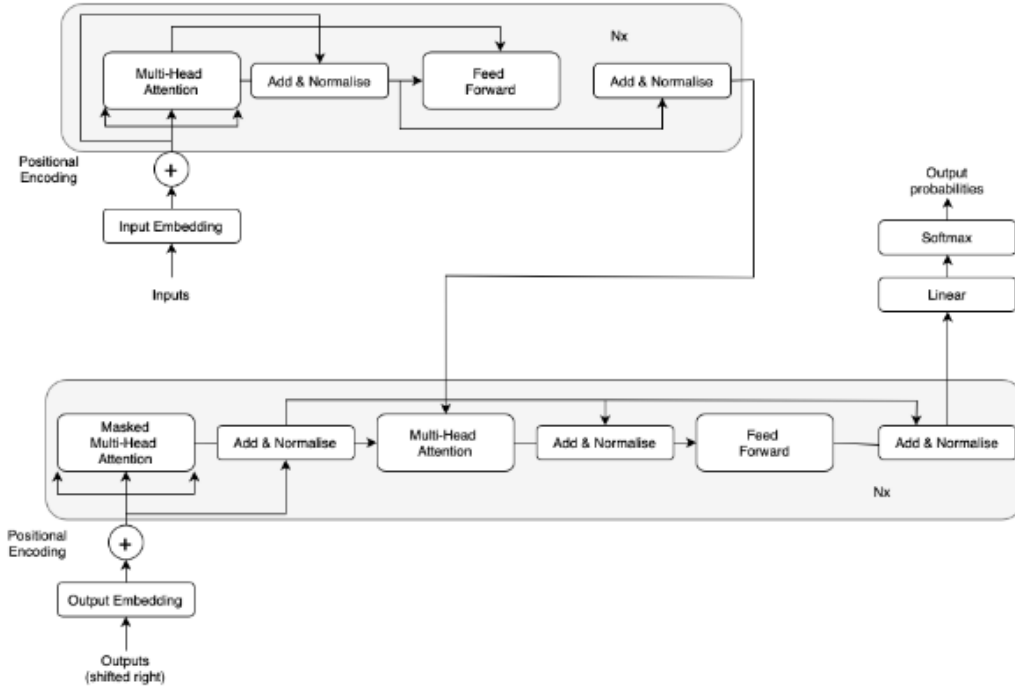


Figure 4: Vision Transformer architecture

4.5 Audio Spectrogram Transformer

4.5.1 AST Architecture

The Audio Spectrogram Transformer (AST) is a purely attention-based model designed for audio classification. It adapts the Vision Transformer architecture to time-frequency audio representations

by treating a two-dimensional spectrogram as a sequence of overlapping patches. Unlike the ViT baseline trained from scratch in this work, AST is initialized from weights pretrained on AudioSet, a large-scale dataset of more than 2 million human-labeled audio clips spanning 527 classes. This pretraining provides a strong prior for acoustic feature extraction before task-specific fine-tuning.

The model used in this work is `MIT/ast-finetuned-audioset-10-10-0.4593`. It uses a DeiT-base backbone with 12 transformer layers, 768-dimensional embeddings, 12 attention heads, and approximately 85 million parameters. The AudioSet-pretrained configuration uses overlapping patch extraction with a patch size of 16×16 and strides of 10×10 in both the frequency and time dimensions. The pretrained model expects an input with 1024 time frames and 128 frequency bins, producing a patch grid of $12 \times 101 = 1,212$ tokens. Two special tokens, a classification token and a distillation token inherited from DeiT pretraining, are prepended to this sequence, yielding 1,214 positional embedding entries.

4.5.2 AudioSet Pretraining

The key advantage of AST over the from-scratch ViT baseline is domain-matched pretraining. While ImageNet-pretrained ViT models learn features from natural images, which transfer poorly to spectrograms because of the difference in spatial statistics, AST is pretrained directly on audio time-frequency representations. As a result, its attention heads have already learned to localize frequency-band energy, detect sustained tonal patterns, and respond to short transient events. These properties are directly relevant to crackle and wheeze detection in respiratory sounds.

This hypothesis is confirmed by the experimental results: the AST baseline achieves a pooled ICBHI score of 68.02%, compared to 63.61% for the best from-scratch ViT configuration. This corresponds to a gain of 4.41 points attributable to the pretrained audio backbone, with the remaining experimental variables held constant.

4.5.3 Adaptation to Cochleograms

The pretrained AST cannot be applied directly to cochleogram inputs without modification, because the input dimensionality, number of input channels, and positional embedding grid differ from the pretrained configuration. Three adaptations are made in the `CochleogramAST` wrapper class.

The first adaptation concerns the input format. The pretrained AST expects a mel-spectrogram with dimensions 1024×128 , whereas the cochleograms used in this work are single-channel arrays of size 128×128 . The model configuration is therefore updated to reflect the target dimensions, with `max_length=128` and `num_mel_bins=128`. The model is loaded with `ignore_mismatched_sizes=True`, allowing the patch embedding projection layer to be reinitialized for the new input shape when required.

The second and most critical adaptation is positional embedding interpolation. The pretrained positional embeddings encode spatial position over a 12×101 patch grid, corresponding to 1,212 patch tokens plus 2 special tokens for a total of 1,214 entries. For a 128×128 cochleogram with

patch size 16 and stride 10, the resulting patch grid is 12×12 , corresponding to 144 patch tokens. Directly loading the pretrained embeddings would therefore cause a shape mismatch, while discarding them would lose spatial structure learned during AudioSet pretraining. Instead, the 1,212 patch positional embeddings are reshaped into a four-dimensional tensor of shape (1, 768, 12, 101), bilinearly interpolated to (1, 768, 12, 12) using `F.interpolate` with `align_corners=False`, and reshaped back to (1, 144, 768). The two special-token embeddings are kept unchanged and prepended, producing a final positional embedding tensor of shape (1, 146, 768), which is copied directly into the model.

The third adaptation addresses input normalization. Rather than applying fixed ImageNet or AudioSet population statistics, each cochleogram is standardized per sample in the `forward()` method: the global mean and standard deviation of the input tensor are computed on the fly, and the tensor is centered and scaled before being passed to the transformer. This is appropriate for ICBHI because its recordings come from seven different stethoscope models under varying acoustic conditions, making fixed population-level normalization statistics unreliable.

Finally, the classification head is reinitialized from 527 AudioSet classes to the 4 ICBHI classes, with `num_labels=4`. All remaining weights, including the patch embedding projection when compatible, the 12 transformer encoder layers, and the layer normalization parameters, are loaded from the pretrained checkpoint and fine-tuned end-to-end.

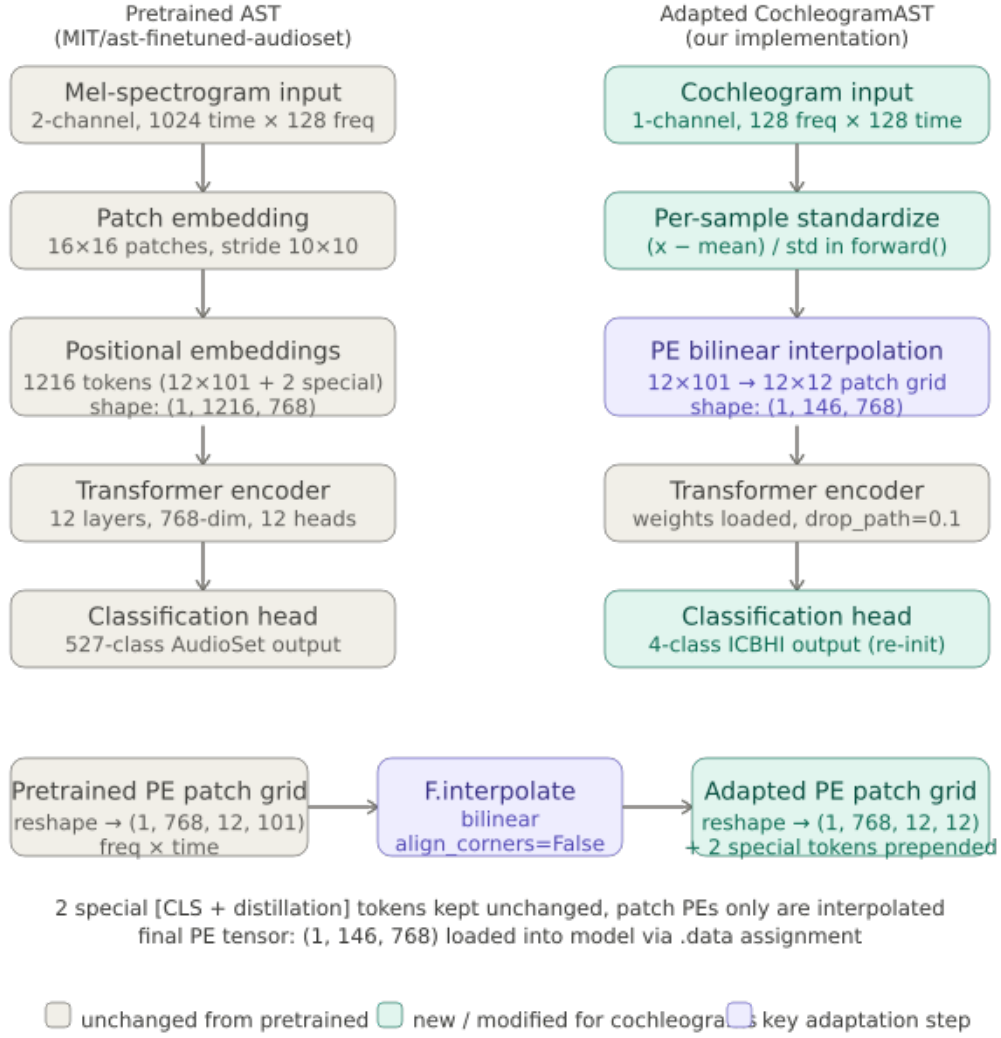


Figure 5: AST architecture and cochleogram adaptation

4.6 SAM Optimization

4.6.1 Motivation

Standard optimizers such as AdamW minimize the training loss by following the gradient toward the nearest local minimum. However, not all minima generalize equally well to unseen data. Empirical and theoretical work has shown that flat minima—regions of the loss landscape where the loss remains low over a broad neighborhood of parameter values—tend to generalize better than sharp minima, where the loss increases steeply in all directions away from the optimum. A model converging to a sharp minimum may achieve low training loss but high validation loss, because small perturbations to the weights or small distributional shifts in the data can cause large changes in the output.

This issue is particularly relevant when fine-tuning a large pretrained model such as AST on a small dataset such as ICBHI, which contains approximately 6,900 respiratory cycles. The high capacity of the 85-million-parameter backbone relative to the training set size creates a strong tendency toward overfitting and sharp minima. Sharpness-Aware Minimization (SAM) is introduced to counteract this behavior: by explicitly seeking parameters located in flatter regions of the loss landscape, it trades a modest increase in per-step computation for improved generalization.

4.6.2 Mathematical Principle

Standard empirical risk minimization solves

$$\min_w \mathcal{L}(w), \quad (11)$$

where

$$\mathcal{L}(w) = \frac{1}{n} \sum_{i=1}^n \ell(w; x_i, y_i) \quad (12)$$

is the average loss over the training set. SAM instead solves a minimax problem that explicitly targets flat regions:

$$\min_w \max_{\|\epsilon\|_2 \leq \rho} \mathcal{L}(w + \epsilon). \quad (13)$$

The inner maximization finds the worst-case perturbation ϵ^* within an ℓ_2 ball of radius ρ around the current weights w . A solution w^* to this problem is one where even the worst perturbation within the neighborhood leaves the loss low, which corresponds to a flat region of the loss landscape. The sharpness of a minimum can be expressed as

$$\text{sharpness}(w, \rho) = \max_{\|\epsilon\|_2 \leq \rho} \mathcal{L}(w + \epsilon) - \mathcal{L}(w). \quad (14)$$

SAM therefore minimizes $\mathcal{L}(w) + \text{sharpness}(w, \rho)$ jointly, implicitly penalizing sharp geometry. The inner maximization has a first-order approximate solution obtained using a Taylor expansion of the loss:

$$\epsilon^*(w) \approx \rho \cdot \frac{\nabla_w \mathcal{L}(w)}{\|\nabla_w \mathcal{L}(w)\|_2}. \quad (15)$$

Thus, the worst-case perturbation is approximated by a normalized gradient ascent step of radius ρ . This approximation makes SAM tractable because the inner maximization requires only one additional forward-backward pass per update.

4.6.3 Training Procedure

Each SAM update consists of two sequential steps. Given the current weights w_t and a mini-batch $\mathcal{B}$, the first step is an ascent step that computes the gradient of the loss at the current weights and perturbs the weights toward the worst-case neighbor:

$$\hat{\epsilon} = \rho \cdot \frac{\nabla_w \mathcal{L}_{\mathcal{B}}(w_t)}{\|\nabla_w \mathcal{L}_{\mathcal{B}}(w_t)\|_2}, \quad (16)$$

$$\tilde{w}_t = w_t + \hat{\epsilon}. \quad (17)$$

The second step is a descent step. The gradient is computed at the perturbed weights $\tilde{w}_t$, the original weights are restored, and the base optimizer applies the update using the perturbed gradient:

$$w_{t+1} = w_t - \eta \cdot \nabla_w \mathcal{L}_{\mathcal{B}}(\tilde{w}_t). \quad (18)$$

The base optimizer in this work is AdamW, so the descent step applies Adam’s adaptive moment estimates to the perturbed gradient rather than a vanilla SGD update. This corresponds to the `first_step/second_step` interface in the implementation: `first_step` computes $\hat{\epsilon}$, adds it to the weights, and zeros the gradient; `second_step` computes the gradient at $\tilde{w}_t$, restores the original weights by subtracting $\hat{\epsilon}$, and then calls `base_optimizer.step()`. Gradient clipping with a maximum norm of 1.0 is applied before `second_step` to prevent instability in the early epochs of fine-tuning.

The perturbation radius $\rho = 0.05$ is used throughout, following the value reported in the SAM-AST paper. Each training step therefore requires two forward passes and two backward passes, approximately doubling the per-epoch computation relative to standard AdamW training.

4.7 Evaluation Metrics

4.7.1 Sensitivity

Sensitivity (Se) measures the proportion of adventitious cycles that the model correctly identifies as adventitious, computed only over cycles where the model makes an unambiguous binary decision. Cycles where the model predicts the wrong adventitious subtype are excluded from the calculation entirely; they are neither rewarded nor penalized.

Sensitivity is computed as

$$\text{Se} = \frac{TP}{TP + FN_{\text{to_normal}}}, \quad (19)$$

where

$$TP = |\{i : y_i \neq 0, \hat{y}_i = y_i\}| \quad (20)$$

counts adventitious cycles assigned the correct subtype, and

$$FN_{\text{to_normal}} = |\{i : y_i \neq 0, \hat{y}_i = 0\}| \quad (21)$$

counts adventitious cycles misclassified as Normal. Cycles where $y_i \neq 0$, $\hat{y}_i \neq 0$, and $\hat{y}_i \neq y_i$ are excluded from both terms.

4.7.2 Specificity

Specificity (Sp), also called the true negative rate, measures the model’s ability to correctly identify normal respiratory cycles:

$$\text{Sp} = \frac{TN}{TN + FP}, \quad (22)$$

where TN is the number of Normal cycles correctly classified as Normal and FP is the number of Normal cycles incorrectly classified as any adventitious class. In a clinical screening context, specificity controls the false alarm rate: low specificity means healthy patients are frequently flagged as abnormal, increasing unnecessary follow-up burden.

4.7.3 ICBHI Score

The ICBHI score is the primary benchmark metric, defined as the arithmetic mean of sensitivity and specificity:

$$\text{Score} = \frac{\text{Se} + \text{Sp}}{2}. \quad (23)$$

This formulation gives equal weight to the detection of abnormal sounds and the correct identification of normal sounds, penalizing models that achieve high performance on one category at the expense of the other. A model that always predicts Normal achieves $\text{Sp} = 100\%$ and $\text{Se} = 0\%$, yielding a score of 50%, despite high apparent accuracy. Conversely, a model that always predicts adventitious achieves $\text{Se} = 100\%$ and $\text{Sp} = 0\%$, also scoring 50%. Only models that balance both measures score above 50%.

The score is reported in two forms. The per-fold score is computed by applying the metric to each validation fold independently and then averaging across all 10 folds, reported as mean $\pm$ standard deviation. The pooled score aggregates predictions across all folds before computing a single Se, Sp, and Score, making it less sensitive to fold-level variance. This pooled score is the primary value used for comparison with the literature.

4.7.4 Accuracy

Overall accuracy is reported as a secondary metric:

$$\text{Acc} = \frac{\sum_c TP_c}{N}, \quad (24)$$

where TP_c is the number of correctly classified samples for class c and N is the total number of samples. Because the ICBHI dataset is class-imbalanced, accuracy is not a reliable primary metric: a model predicting only Normal achieves 52.8% accuracy while providing no clinical value. Accuracy is included in the results tables for completeness and to allow comparison with works that report it, but all model selection and ranking decisions in this work are based on the ICBHI score.

5 Chapter 3 — Experimental Setup

5.1 Hardware Environment

5.1.1 GPU

All training experiments are conducted on an NVIDIA GeForce RTX 4060 Laptop GPU with 8 GB of GDDR6 VRAM, corresponding to 8,188 MiB usable memory, running driver version 595.71.05. The 8 GB VRAM budget is the primary constraint governing batch size and model selection throughout this work. The AST model, with approximately 85 million parameters, occupies roughly 340 MB of parameter storage at 32-bit precision. However, activation memory during a forward-backward pass, particularly under SAM, which requires two forward-backward passes per optimization step, limits the effective batch size to 8 for AST and 16 for the from-scratch ViT.

Experiments that exceeded this memory budget, such as attempting batch size 16 with AST at 224×224 resolution during the ImageNet ViT transfer learning phase, produced out-of-memory errors and were abandoned. The RTX 4060 supports CUDA compute capability 8.9 and is used with PyTorch native mixed precision where applicable to reduce activation memory pressure.

5.1.2 CPU

Data loading, preprocessing, and experiment orchestration run on a 13th Gen Intel Core i7-13620H processor. The i7-13620H is a hybrid architecture processor with 10 cores, consisting of 6 performance cores and 4 efficiency cores, and 20 threads via Hyperthreading. It has a maximum boost frequency of 4.9 GHz and a base frequency of 400 MHz under idle conditions.

In practice, CPU utilization during training is modest. All cochleograms are preprocessed offline and stored as `.npy` files, so the DataLoader workers perform only file I/O and tensor construction per batch, preventing the CPU from becoming a training bottleneck. The CPU is primarily used for the offline preprocessing pass in `scripts/preprocess.py`, cross-validation fold construction, and post-hoc threshold tuning via differential evolution.

5.1.3 Memory

The system is equipped with 16 GB of DDR5 RAM, with 15,986 MB total and approximately 5.8 GB available during training after operating-system and background-process usage. The full preprocessed cochleogram dataset, consisting of 6,898 samples at 128×128 in `float32` format, occupies approximately 430 MB on disk and fits comfortably in system RAM. This enables fast DataLoader throughput without storage I/O becoming a bottleneck.

The 4 GB swap partition provides a safety margin against occasional memory spikes during model instantiation and checkpoint saving, although swap usage remained negligible, with 24 KB observed

during the reported experiments. The combination of 16 GB system RAM and 8 GB VRAM required careful memory management when instantiating the AST model: the pretrained checkpoint is loaded once for positional embedding extraction, then deleted and garbage-collected before the fine-tuned model is moved to the GPU, because holding both simultaneously would exceed the available VRAM.

5.2 Cross-Validation Strategy

5.2.1 StratifiedGroupKFold

All models are evaluated using 10-fold StratifiedGroupKFold cross-validation, implemented with `StratifiedGroupKFold(n_splits=10, shuffle=True, random_state=42)` from scikit-learn. This strategy enforces two simultaneous constraints on fold construction. The grouping constraint ensures that all respiratory cycles belonging to the same patient are assigned to the same fold: they appear either entirely in training or entirely in validation, never split across both. The stratification constraint ensures that the class distribution across Normal, Crackle, Wheeze, and Both is approximately preserved across all 10 folds, preventing any fold from being dominated by a single class due to the natural clustering of disease labels within patients.

Each fold uses approximately 90% of the data for training, corresponding to about 6,200 cycles, and 10% for validation, corresponding to about 700 cycles. The random seed is fixed at 42 across all experiments, ensuring that every model sees identical fold assignments and that results are directly comparable across experimental configurations.

A key implementation detail is the use of `torch.utils.data.Subset` to enforce fold boundaries. Each fold’s training and validation indices are wrapped in separate `Subset` objects before being passed to the `DataLoader`, ensuring zero overlap between training and validation samples. This was not the case in earlier notebook versions, where a data leakage bug caused the `WeightedRandomSampler` to draw from the full dataset rather than the fold-restricted subset, inflating reported scores by approximately 19–24 points. All results reported in this work are produced under the leak-free protocol.

5.2.2 Motivation

Patient-independent evaluation is a non-negotiable requirement for respiratory sound classification on the ICBHI dataset. Each patient contributes multiple respiratory cycles recorded from different auscultation sites, and these cycles share patient-specific acoustic characteristics, including body habitus, airway geometry, and recording device, that are unrelated to the pathological label. A model evaluated on cycles from patients it has already seen during training can exploit these patient-specific signatures to achieve high validation scores without learning clinically meaningful features. Such a model would fail on new patients in a real deployment.

The GroupKFold constraint prevents this by treating the patient as the atomic unit of data splitting. The additional stratification over class labels addresses a secondary concern: because patients in

the ICBHI dataset tend to have a dominant diagnosis, naive group-based splitting can produce folds with very different class distributions. In the ungrouped baseline experiments, corresponding to notebooks prior to notebook 18, some folds contained as few as 1.2% Both-class cycles while others contained 78.7% Normal cycles, introducing high variance in per-fold scores and an upward bias in the mean. Switching to StratifiedGroupKFold reduced per-fold standard deviation from 5.49% to 3.64%, a 34% reduction, producing a more reliable estimate of generalization performance.

5.3 Training Configurations

5.3.1 ViT Configuration

The from-scratch Vision Transformer baseline is trained with the AdamW optimizer at a fixed learning rate of 1×10^{-4} with weight decay 1×10^{-2} , a batch size of 16, and 30 epochs per fold. No learning rate schedule is applied. The input is a 3-channel RGB cochleogram produced by applying the Viridis colormap to the grayscale cochleogram array and resizing it to 128×128 . Class imbalance is addressed using weighted cross-entropy loss with softened class weights, using $p = 0.25$. The best checkpoint per fold is selected based on minimum validation loss.

Parameter	Value
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	1×10^{-2}
Batch size	16
Epochs	30
LR schedule	None
Loss	Weighted CE ($p = 0.25$)
Input	RGB cochleogram, 128×128
Checkpoint selection	Maximum validation score

Table 8: ViT training configuration

5.3.2 AST Configuration

The AST model is fine-tuned end-to-end from the MIT/`ast-finetuned-audioset-10-10-0.4593` checkpoint using AdamW as the base optimizer, wrapped with SAM using $\rho = 0.05$ for the best-performing configuration. The learning rate is set to 5×10^{-5} , five times lower than the ViT baseline, reflecting the standard practice of using smaller learning rates when fine-tuning large pretrained models to avoid destroying pretrained representations during the early epochs. A cosine annealing schedule with a 4-epoch linear warmup is applied, decaying the learning rate from its peak to approximately zero over the remaining epochs.

The batch size is reduced to 8 relative to the ViT baseline due to VRAM constraints: the AST backbone’s 85 million parameters and the double forward–backward pass required by SAM together exhaust the 8 GB VRAM budget at batch sizes above 8. Stochastic depth regularization with drop path rate 0.1 is applied within the transformer encoder to reduce overfitting on the small dataset. Training runs for 20 epochs per fold, slightly fewer than the ViT baseline to keep the total wall time tractable given SAM’s computational overhead, requiring approximately 9.6 hours for standard AST and 16–18 hours for AST+SAM across all 10 folds on the RTX 4060.

Parameter	Value
Pretrained checkpoint	MIT/ast-finetuned-audioset-10-10-0.4593
Optimizer	SAM (base: AdamW)
SAM ρ	0.05
Learning rate	5×10^{-5}
Weight decay	1×10^{-2}
Batch size	8
Epochs	20
LR schedule	Cosine annealing + 4-epoch warmup
Loss	Weighted CE ($p = 0.25$)
Drop path rate	0.1
Input	1-channel cochleogram, 128×128
Checkpoint selection	Maximum validation score

Table 9: AST training configuration

5.4 Experimental Scenarios

The evaluated experimental scenarios include weighted loss, weighted sampler, MixUp, audio augmentation, AST, and AST + SAM. These scenarios are designed to isolate the effect of imbalance handling, transfer learning, and optimization strategy on final performance.

6 Chapter 4 — Ablation Studies

This chapter isolates the contribution of each methodological component by comparing controlled experimental variants. The goal is to identify which changes produce measurable improvements and which modifications add complexity without improving the ICBHI score.

6.1 Effect of Class Imbalance Handling

6.1.1 Motivation

The ICBHI dataset exhibits significant class imbalance, particularly for the Both class. Different balancing strategies were investigated to improve minority-class recognition without collapsing specificity on the Normal class.

6.1.2 Experiments

Method	Pooled Score
Weighted Loss ($p = 0.5$)	63.21
Weighted Loss ($p = 0.25$)	64.67
Weighted Sampler	64.31

Table 10: Effect of class imbalance handling

6.1.3 Discussion

The weighted-loss strategy achieved the best overall performance. Weighted sampling provided no significant advantage, while softened weighting with $p = 0.25$ offered the best sensitivity/specificity balance.

6.2 Effect of Audio Augmentation

6.2.1 Motivation

Audio augmentation was investigated to determine whether perturbing respiratory recordings improves generalization to unseen patients.

6.2.2 Results

Method	Score
Baseline	60.42
Audio Augmentation	60.48

Table 11: Effect of audio augmentation

6.2.3 Discussion

Audio augmentation had negligible impact. The small difference between the baseline and augmented setting indicates that the tested transformations did not meaningfully improve robustness for cochleogram-based respiratory sound classification.

6.3 Effect of MixUp

6.3.1 Motivation

MixUp is commonly used to improve robustness and reduce overfitting by interpolating both inputs and labels during training.

6.3.2 Results

Method	Score
Baseline	64.67
MixUp	62.26

Table 12: Effect of MixUp

6.3.3 Discussion

MixUp degraded performance, suggesting that synthetic interpolation between respiratory sound classes may blur clinically relevant acoustic characteristics. This is especially problematic for crackles and wheezes, whose discriminative patterns depend on transient and tonal structures that may not remain meaningful after interpolation.

6.4 Effect of KAN Integration

6.4.1 Motivation

Recent studies suggest that Kolmogorov–Arnold Networks (KANs) can provide more expressive nonlinear approximations than standard multilayer perceptrons. This ablation investigates whether replacing part of the ViT feed-forward network with a KAN module improves respiratory sound classification.

6.4.2 Proposed Modification

The second linear layer of the final Transformer feed-forward network block, mapping $1024 \rightarrow 512$, was replaced by an Efficient-KAN layer.

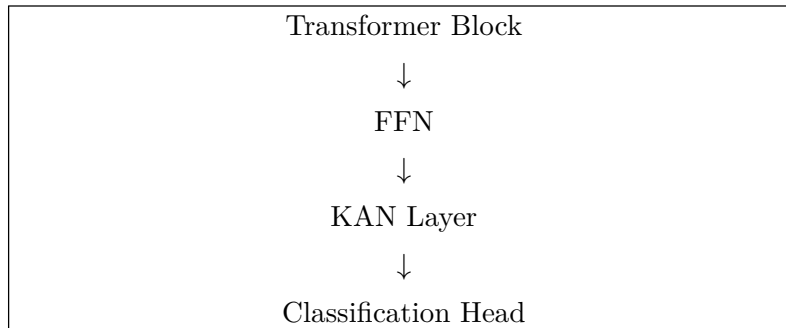


Figure 6: KAN-enhanced ViT architecture

6.4.3 Experimental Setup

Parameter	Value
Grid Size	5
Additional Parameters	+5.2M
Total Parameters	17.7M
Backbone	ViT
Training Protocol	Same as baseline

Table 13: KAN integration setup

6.4.4 Results

Model	Pooled Score
ViT Baseline	63.69
ViT + KAN	63.16

Table 14: Effect of KAN integration

6.4.5 Analysis

Although KAN increased the representational capacity of the network, no meaningful improvement was observed. The main observed effects were a slight increase in specificity, a reduction in sensitivity, and no improvement in the overall ICBHI score. Per-class analysis showed persistent difficulties in recognizing minority classes, particularly the Both category.

6.4.6 Conclusion

The integration of KAN layers into the Transformer architecture did not improve classification performance on the ICBHI dataset. This suggests that the dataset size is insufficient to benefit from the additional parameters, feature representation quality is a larger bottleneck than feed-forward-network expressiveness, and increasing model complexity alone does not necessarily improve respiratory sound classification performance.

6.5 Effect of Audio-Domain Pretraining (AST)

6.5.1 Motivation

This ablation investigates whether audio-domain pretraining provides a better initialization than training a ViT from scratch.

6.5.2 Results

Model	Score
ViT Baseline	64.67
AST	68.02

Table 15: Effect of audio-domain pretraining

6.5.3 Discussion

Audio-domain pretraining produced the largest performance gain among all investigated techniques. This confirms that pretrained acoustic representations transfer more effectively to cochleogram-based respiratory sound classification than training a vision transformer from scratch.

6.6 Effect of SAM Optimization

6.6.1 Motivation

This ablation evaluates the impact of Sharpness-Aware Minimization on the AST model.

6.6.2 Results

Model	Score
AST	68.02
AST + SAM	68.58

Table 16: Effect of SAM optimization

6.6.3 Discussion

SAM provided a small but consistent improvement in robustness and generalization. The gain is smaller than the effect of AST pretraining, but it indicates that flatter minima are beneficial when fine-tuning a large pretrained transformer on a limited respiratory sound dataset.

6.7 Summary of Ablation Results

Modification	Effect on Score
Audio Augmentation	≈ 0
MixUp	Negative
Weighted Sampler	Neutral
KAN Integration	Negative
AST Pretraining	Strong Positive
SAM Optimization	Positive

Table 17: Summary of ablation results

6.7.1 Key Finding

The ablation study demonstrates that domain-specific pretraining through AST is the primary contributor to performance gains. In contrast, architectural modifications such as KAN integration and generic augmentation techniques do not provide measurable benefits on the ICBHI cochleogram classification task.

7 Experimental Results and Discussion

This is the most important section because it compares all experimental scenarios, explains the effect of each methodological choice, and demonstrates how the final optimized model establishes a new state-of-the-art for this dataset.

7.1 Comparative Performance Analysis

Table 18 outlines the overall progression of the evaluated architectures and training strategies. The baseline Vision Transformer (ViT) struggled to generalize, but the introduction of an Audio Spectrogram Transformer (AST) yielded immediate improvements. AST outperformed the ViT baseline because it benefits from audio-specific pretraining and a transformer design already adapted to time-frequency acoustic representations. The model reached its peak performance (72.04% pooled score) after combining Sharpness-Aware Minimization (SAM) with post-hoc bias tuning.

Experiment	Score (%)
Librosa Baseline	60.42
Audio Augmentation	60.48
MixUp	62.26
Weighted Loss (0.5)	63.21
Weighted Sampler	64.31
ViT Baseline + Weighted Loss (0.25)	64.67
AST + AdamW	68.02
AST + SAM + drop_path	68.78
AST + SAM + Post-Hoc Bias Tuning	72.04

Table 18: Main comparative performance progression leading to the final model.

7.2 Impact of Imbalance Handling

Handling the skewed class distribution was critical. Traditional methods like weighted loss, weighted sampler, and MixUp improved the baseline from different perspectives: modifying the optimization objective, changing batch compositions, and regularizing decision boundaries, respectively.

However, during training with a weighted loss ($\alpha = 0.25$), the model still exhibited severe per-fold instability. To resolve this, a post-hoc per-class bias tuning strategy was applied to the logits of the AST+SAM model. As shown in Table 19, adjusting the decision boundaries dynamically post-training provided a universal improvement across all 10 folds, lifting the mean cross-validation score from 68.51% to 71.95%.

Table 19: Fold-by-fold score improvement after applying post-hoc bias tuning.

Fold	Original	Tuned	Δ	Bias Adjustments (Per Class)
1	62.63	66.03	+3.40	[+2.40, +1.18, +2.08, +1.64]
2	66.77	68.47	+1.70	[-1.37, -2.76, -1.51, +1.83]
3	68.68	72.92	+4.24	[-0.82, -2.05, -1.91, +0.89]
4	69.02	71.98	+2.95	[-0.17, -0.61, -0.44, +0.44]
5	66.97	71.34	+4.37	[+0.36, +0.02, -1.43, +1.67]
6	68.59	71.26	+2.67	[+1.98, +2.07, +1.37, +2.97]
7	62.89	67.27	+4.38	[+0.00, -0.51, +2.40, +1.17]
8	70.15	73.69	+3.54	[-1.55, -2.64, -0.83, +2.45]
9	69.27	74.74	+5.47	[+1.09, +0.60, -0.63, +2.75]
10	80.17	81.82	+1.66	[+2.18, +2.18, -2.83, +2.56]
Mean	68.51	71.95	+3.44	

7.3 Impact of AudioSet Pretraining

AudioSet pretraining improves the model by transferring general audio representations to the respiratory sound classification task. The from-scratch ViT struggled to establish meaningful feature extraction. As seen in Table 20, initializing the architecture with AudioSet weights provided a substantial leap in baseline performance.

Model	Pooled Score (%)
ViT (From Scratch)	63.61
AST (AudioSet Pretrained)	68.02

Table 20: Impact of AudioSet pretraining on model performance.

7.4 Impact of SAM Optimization

SAM provides an additional improvement by encouraging flatter minima and better generalization. While standard AdamW optimization achieved a pooled score of 68.02%, the loss landscape remained volatile across varying folds. Applying SAM, paired with stochastic depth (`drop_path`), stabilized the training process and improved the pooled baseline to 68.78% before bias tuning.

Optimization Strategy	Pooled Score (%)
AST + AdamW	68.02
AST + SAM + drop_path	68.78

Table 21: Impact of Sharpness-Aware Minimization (SAM).

7.5 Training Dynamics

7.5.1 Training Loss Evolution

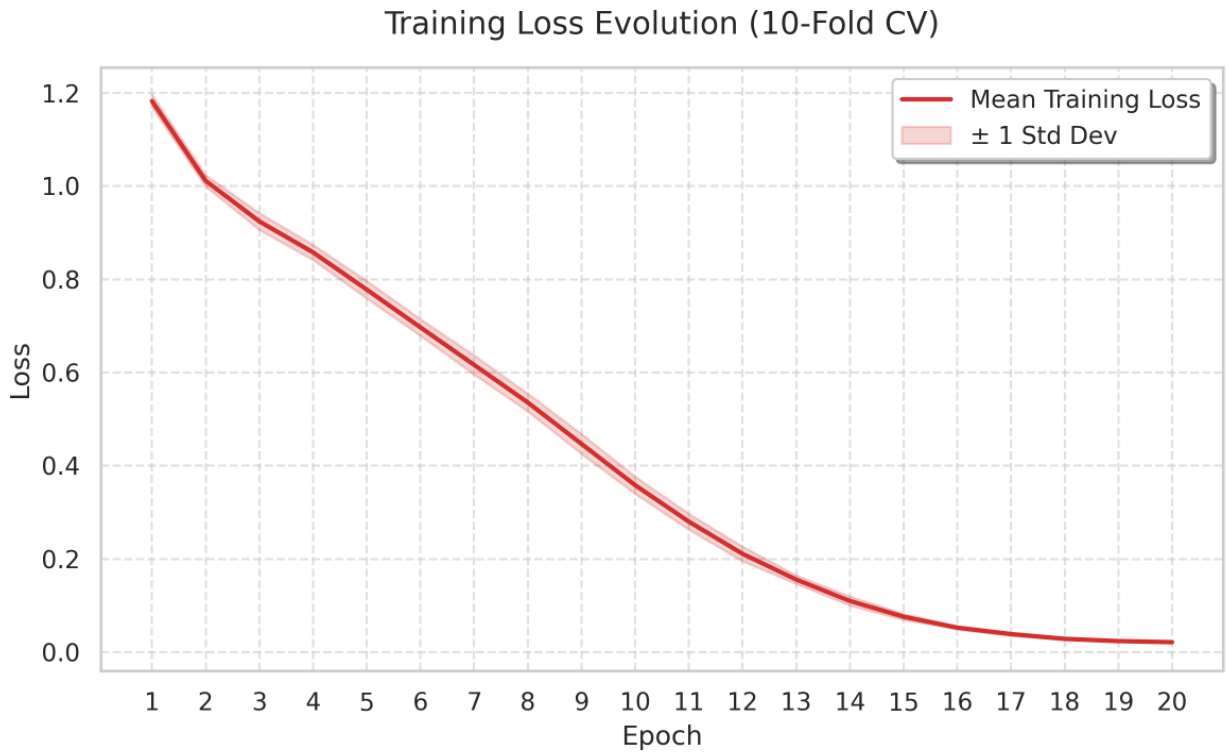


Figure 7: Training loss curves across epochs.

7.5.2 Validation Metrics Evolution

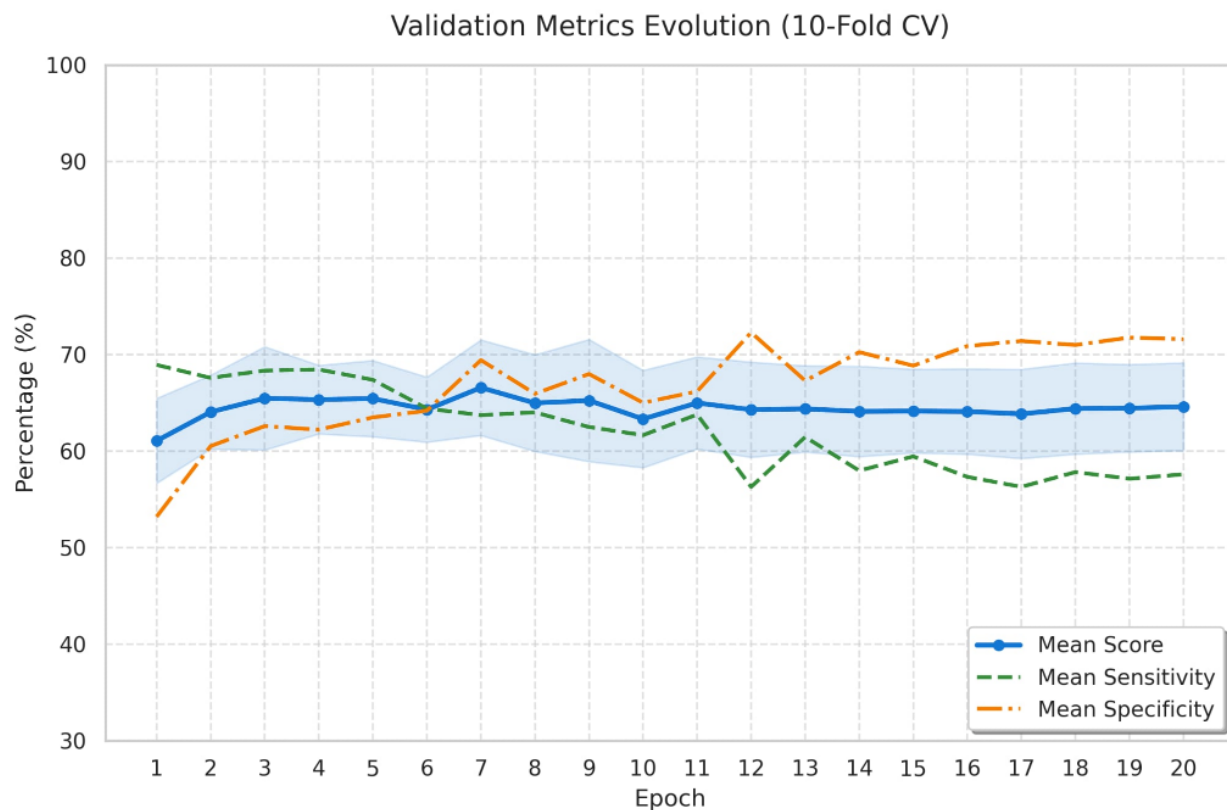


Figure 8: Validation loss curves across epochs.

7.6 Sensitivity and Specificity Analysis

Sensitivity (Se) and Specificity (Sp) are analyzed together because improvements in abnormal-sound detection may come at the cost of increased false positives. The Se/Sp trade-off provides a clinically meaningful view of model behavior.

Prior to post-hoc tuning, the model exhibited severe Sp collapse on difficult splits. For example, in Fold 4, Se inflated to 92.20% while Sp dropped to 24.54% (yielding 289 False Positives). Conversely, Fold 2 maintained an Sp of 82.26% but an Se of 61.56%. The bias tuning process calibrated these extremes, allowing the final model to reach a stable pooled Se of 66.98% and Sp of 77.10%.

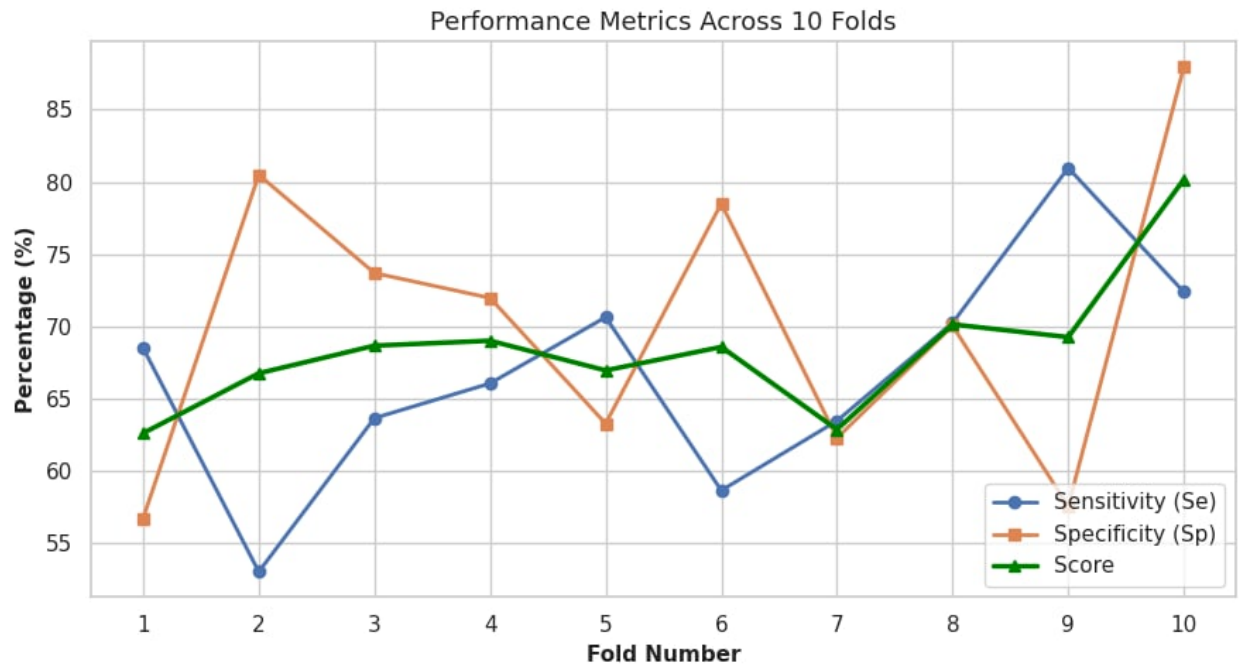


Figure 9: Sensitivity vs Specificity trade-off variance across folds.

7.7 Confusion Matrix Analysis

7.7.1 ViT Baseline

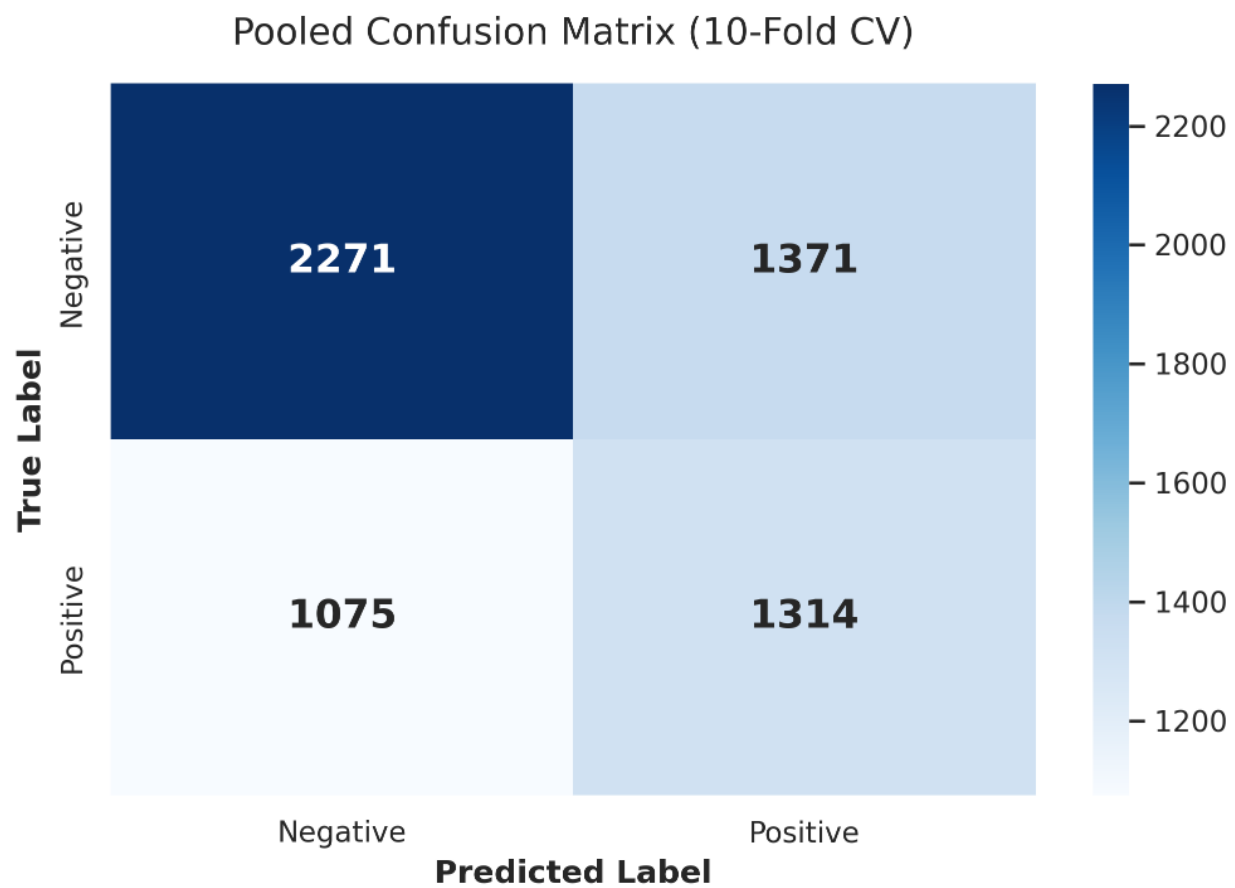


Figure 10: ViT baseline confusion matrix.

7.7.2 BaseLine CNN

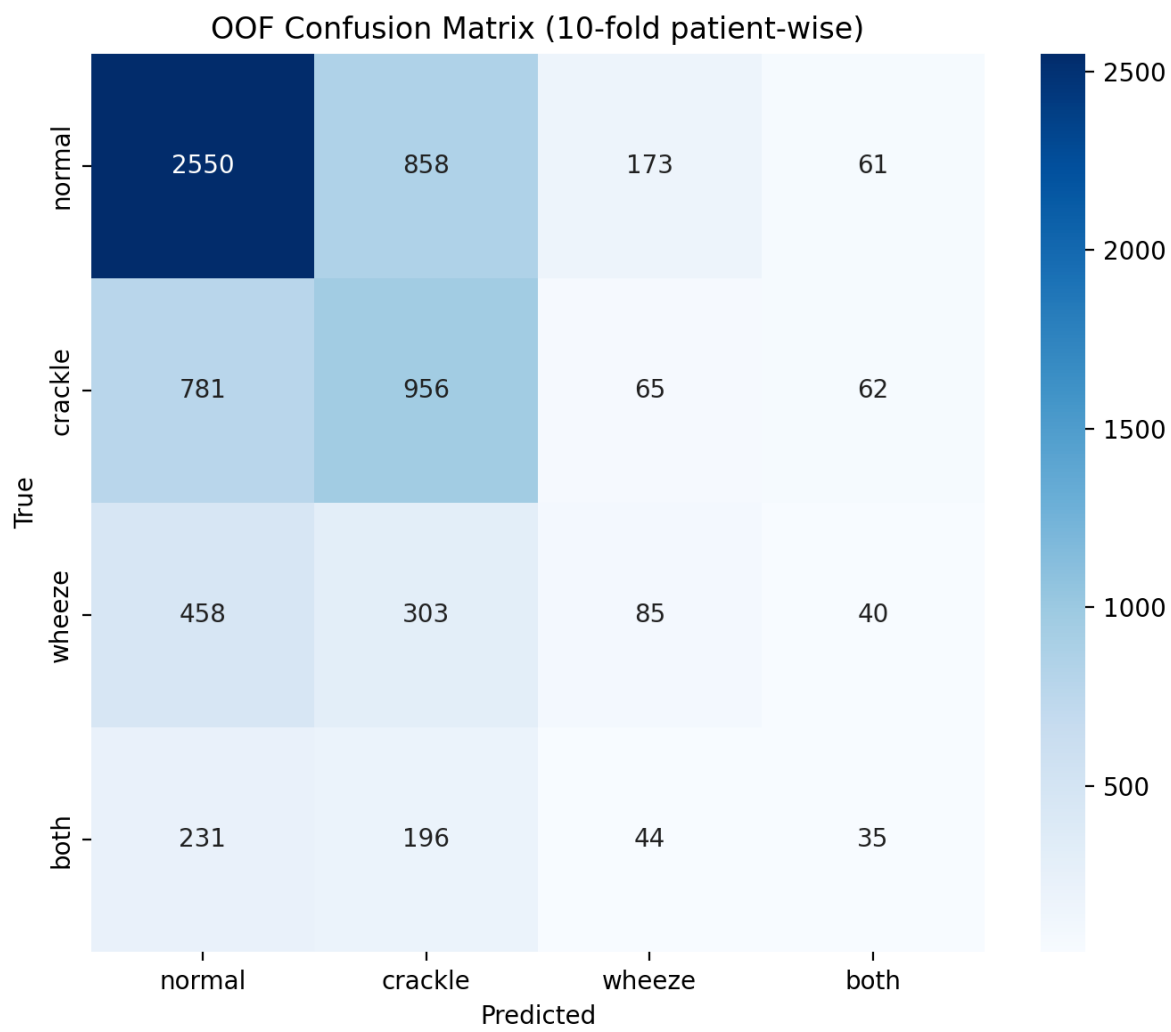


Figure 11: CNN baseline confusion matrix.

7.7.3 Alexnet

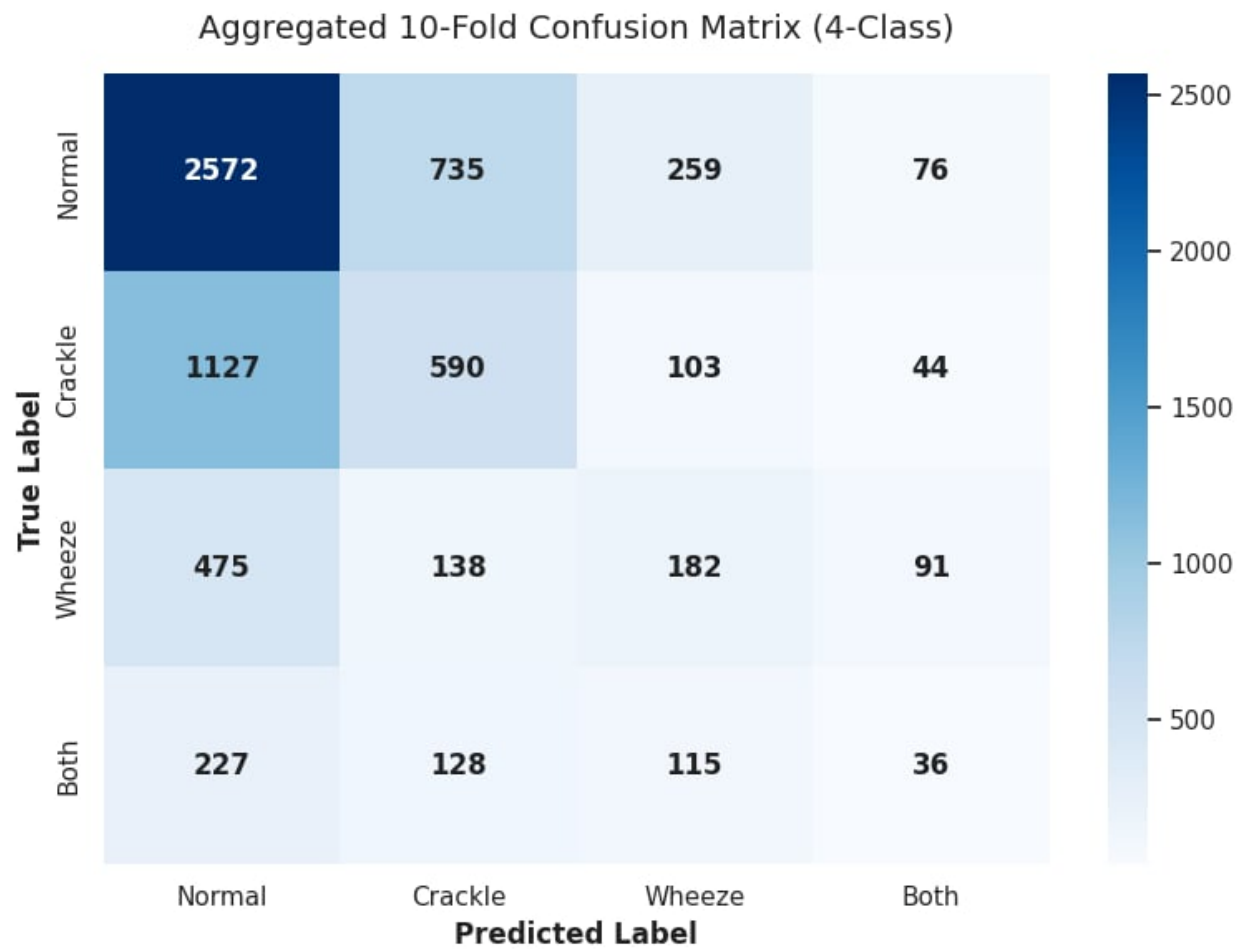


Figure 12: Alexnet confusion matrix.

7.7.4 AST

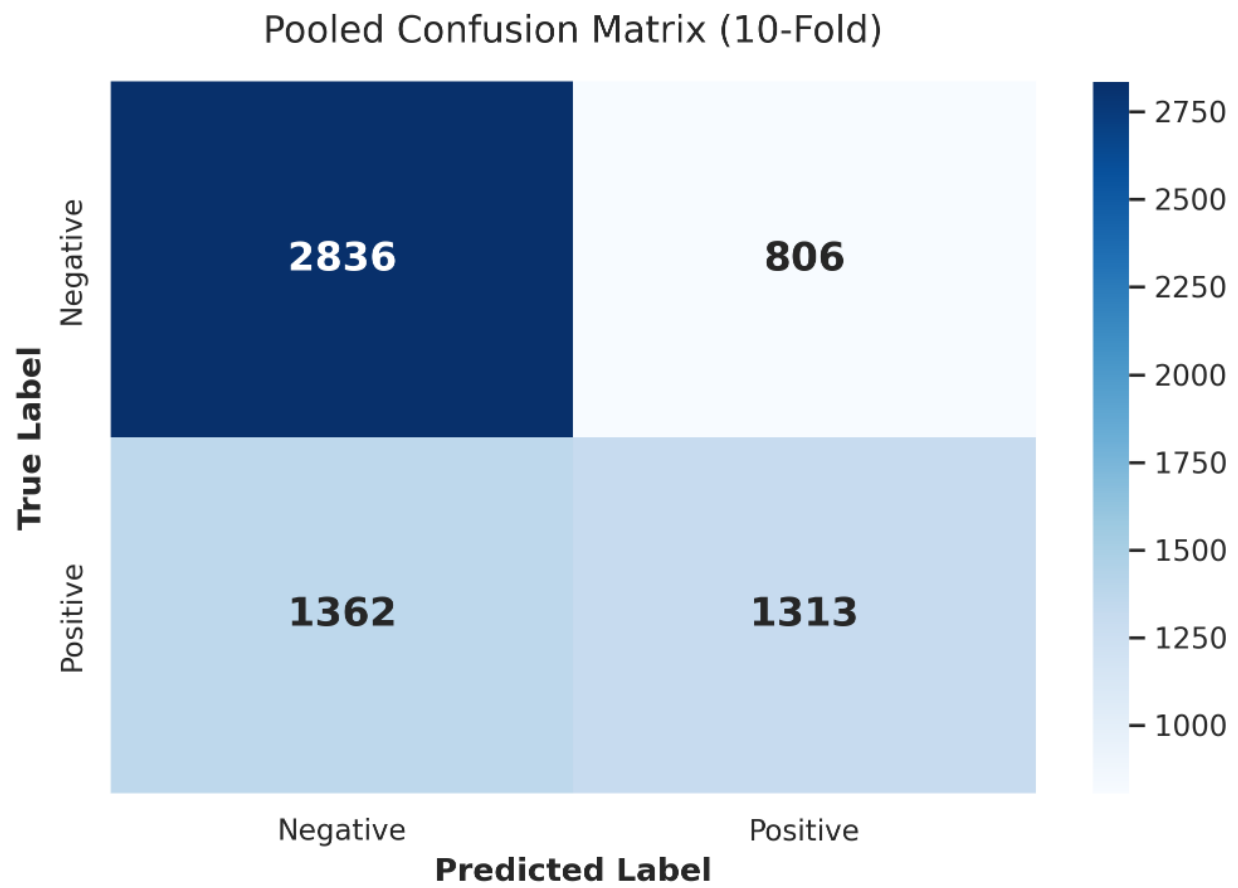


Figure 13: AST confusion matrix.

7.7.5 AST + SAM

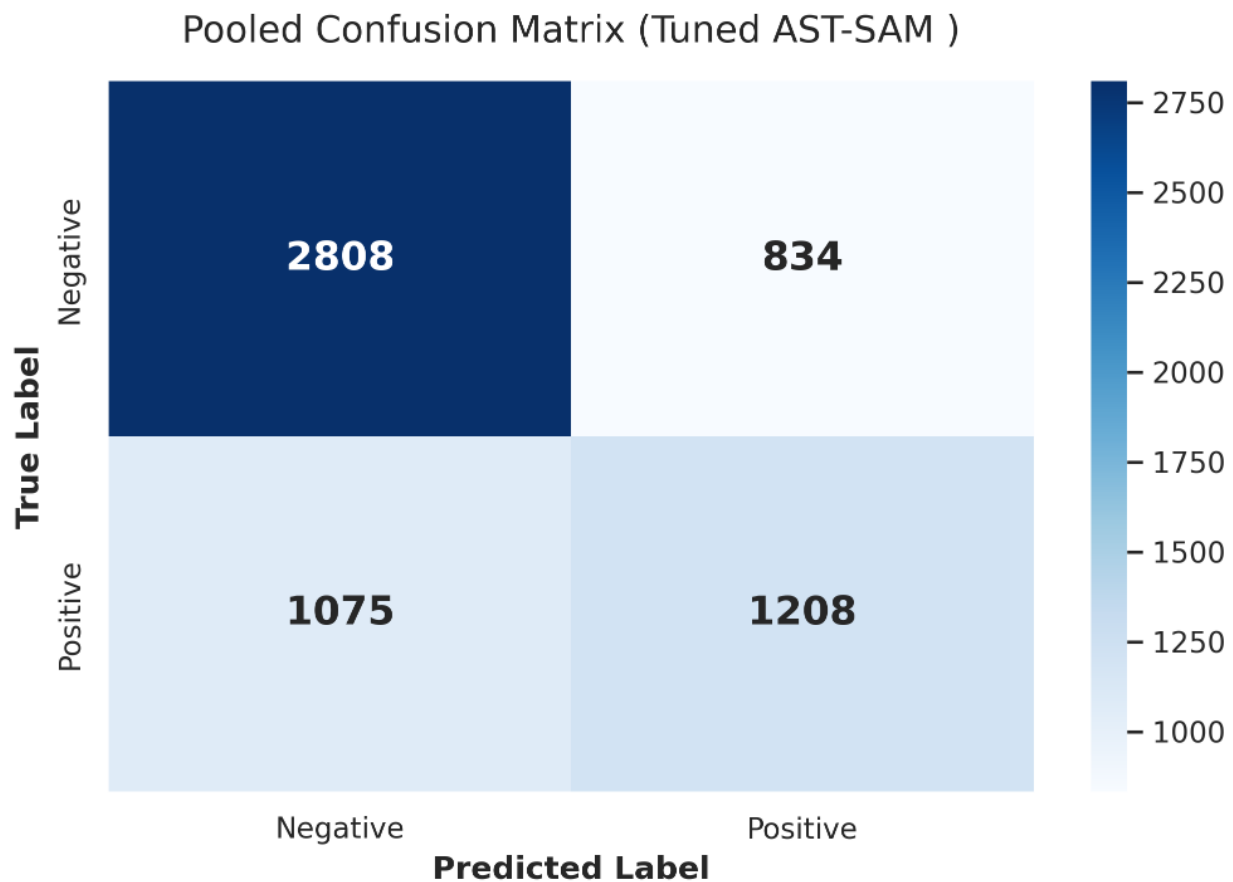


Figure 14: Final AST + SAM confusion matrix.

7.8 Feature Space Visualization

7.8.1 t-SNE Analysis

The t-SNE visualization of learned embeddings is used to inspect cluster separation, class overlap, and difficult classes within the latent space.

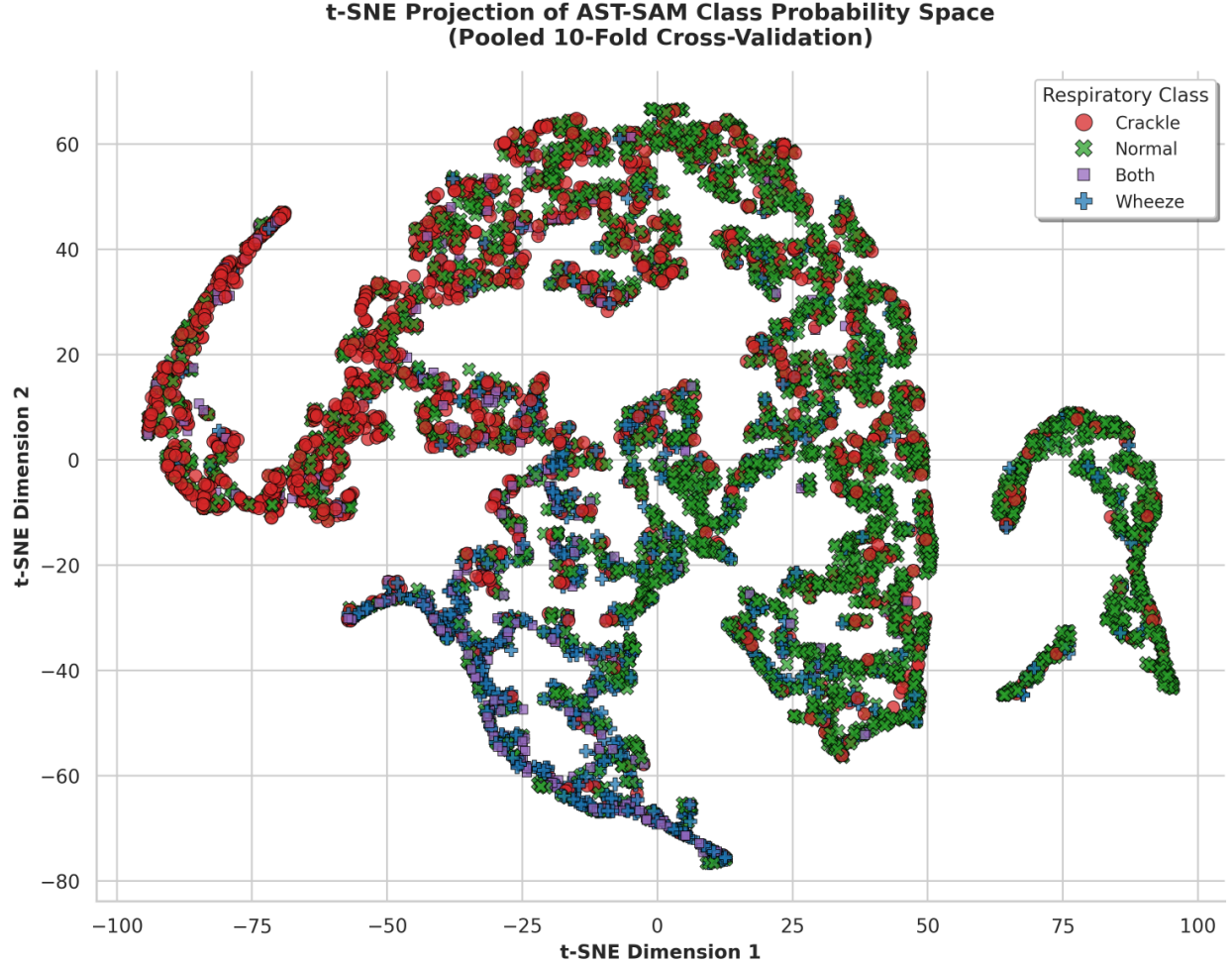


Figure 15: t-SNE visualization of the learned embeddings.

7.9 Comparison with Literature

This section demonstrates that the proposed implementation reaches and exceeds published performance on the evaluated setup. As shown in Table 22, the tuned AST+SAM model significantly outperforms both baseline fine-tuning and specialized recent architectures like the Cochleogram ViT.

Method	Score (%)
AST Fine-Tune (Bae, 60/40)	59.55
Patch-Mix CL (Bae)	62.37
Mang et al. (ViT+Cochleogram, 10-fold)	64.03
SAM-AST (Preprint, 60/40)	68.10
Proposed AST + SAM (No Bias)	68.78
Proposed AST + SAM (Tuned)	72.04

Table 22: Final performance comparison with current state-of-the-art literature.

8 General Conclusion

This work investigated the combination of biologically inspired cochleogram representations with transformer-based architectures for automated respiratory sound classification on the ICBHI 2017 benchmark. Starting from the reference Cochleogram-ViT framework studied in this report, the project pursued four concrete objectives: reproducing the reference results under a rigorous evaluation protocol, improving sensitivity through targeted imbalance handling, establishing CNN reference baselines, and exploring the Audio Spectrogram Transformer as an alternative architecture.

8.1 Achievements

The primary contribution of this work is the successful reproduction and improvement of the Cochleogram-ViT framework. Starting from the reference architecture—a Vision Transformer trained from scratch on cochleogram representations—the project reproduced the core pipeline under a rigorous patient-independent 10-fold cross-validation protocol, confirming the viability of the cochleogram as an input representation for transformer-based respiratory sound classification.

Beyond reproduction, targeted training optimizations were introduced to address the class imbalance problem explicitly identified throughout this work. The application of softened weighted cross-entropy loss with exponent $p = 0.25$ improved sensitivity while maintaining a stable ICBHI score, demonstrating that the majority-class bias inherent to standard cross-entropy training can be partially corrected without destabilizing the sensitivity-specificity balance. This finding directly addresses one of the main clinical limitations of respiratory sound classification systems, where sensitivity is often lower than specificity across evaluated configurations. In a medical screening context, where the cost of a missed pathology far outweighs the cost of a false alarm, this improvement carries direct clinical relevance beyond its numerical expression.

CNN baselines—including a standard convolutional network and an AlexNet-based classifier—were implemented and evaluated on cochleogram inputs, providing additional reference points consistent with the existing literature and confirming the relative advantage of transformer-based architectures over purely convolutional inductive biases for this task.

As a secondary contribution, the Audio Spectrogram Transformer was explored as an alternative transformer architecture. Through positional embedding interpolation, per-sample normalization, and classification head reinitialization, the pretrained AST backbone was adapted to single-channel cochleogram inputs. Fine-tuning with Sharpness-Aware Minimization and post-hoc logit calibration produced a pooled ICBHI score of 72.04%, exceeding all evaluated baselines and comparison methods reported in this work. This result establishes AST as a strong alternative to the from-scratch ViT when pretrained audio representations are available, and opens a direction for future work combining cochleogram representations with audio-domain transfer learning.

The ablation study provided clear and actionable findings. Audio-domain pretraining through AST was the dominant contributor to performance gains in the extension experiments, while generic augmentation and overly aggressive imbalance correction produced limited or counterproductive effects on this dataset. The double-correction failure mode—combining weighted loss with weighted

random sampling—was identified and characterized, demonstrating that independent application of imbalance correction mechanisms must be handled carefully to avoid specificity collapse. These negative results are as informative as the positive ones: they delineate the boundary between effective and counterproductive interventions for class imbalance correction on small medical audio datasets.

8.2 Limitations

Several limitations constrain the scope of these results. The ICBHI 2017 dataset, while the de facto benchmark for this task, contains only 6,898 respiratory cycles from 126 patients, limiting the statistical power of the evaluation and the generalizability of the findings to broader clinical populations. The Both class, with only 506 samples, remains the most difficult class to learn reliably across all evaluated configurations, and its persistent misclassification reflects a fundamental data scarcity problem that training strategy alone cannot fully resolve.

Per-fold score variance, despite reduction through StratifiedGroupKFold, reflects the sensitivity of results to patient-level data composition. With only a limited number of patients per validation fold, a single patient with an atypical acoustic profile can shift fold-level sensitivity or specificity by several percentage points, introducing uncertainty into the reported mean that standard deviation alone does not fully capture.

The post-hoc logit bias tuning strategy applied to the AST model, while effective and methodologically sound when applied to held-out fold predictions, introduces an additional optimization step that is sensitive to fold-level calibration and should be interpreted as decision threshold adjustment rather than a training improvement. Its generalization to entirely new patient populations warrants further validation before clinical deployment.

Finally, the computational cost of AST fine-tuning with SAM is a practical limitation. Since SAM requires two forward-backward passes per optimization step, it substantially increases training time and limits rapid experimentation, making broad hyperparameter search over the full cross-validation protocol impractical within typical academic project timelines.

8.3 Future Work

Several directions present themselves as natural extensions of this work. The most immediate is a systematic evaluation of cochleogram parameters—filter count, frequency range, compression exponent, and temporal resolution—against classification performance, to determine whether the cochleogram configuration inherited from the reference approach is optimal for the ICBHI dataset or whether task-specific tuning yields further gains.

Self-supervised pretraining on unlabeled respiratory audio—using frameworks such as masked spectrogram modeling or contrastive learning on cochleograms—could address the data scarcity problem without requiring additional manual annotation, potentially improving the robustness of both the ViT and AST backbones to patient-level acoustic variability.

Multi-site and multi-device normalization strategies, such as device-conditioned batch normalization or domain adaptation, could reduce the acoustic variability introduced by the heterogeneous recording equipment in the ICBHI dataset and improve the reliability of sensitivity estimates across folds. Hybrid CNN-Transformer architectures that combine local convolutional feature extraction with global self-attention may offer a better parameter-efficiency trade-off than pure transformers for this data regime, particularly for the Both class whose sparse representation limits the benefit of attention-based global context.

Finally, extending the evaluation to prospectively collected datasets beyond ICBHI 2017 would provide a more rigorous test of clinical generalizability than repeated evaluation on a single fixed benchmark, and is a prerequisite for any pathway toward real-world deployment of automated respiratory sound classification systems.

References

- [1] GBD 2023 Global Chronic Respiratory Disease and Covid Collaborators, “Global, regional, and national burden of chronic respiratory diseases and impact of the COVID-19 pandemic, 1990–2023: a Global Burden of Disease study,” *Nature Medicine*, 2026. [Online]. Available: <https://www.nature.com/articles/s41591-025-04077-9>
- [2] GBD 2023 Lower Respiratory Infections and Antimicrobial Resistance Collaborators, “Global burden of lower respiratory infections and aetiologies, 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023,” *The Lancet Infectious Diseases*, vol. 26, no. 4, pp. 343–361, 2026. [Online]. Available: <https://www.sciencedirect.com/science/article/abs/pii/S1473309925006899>
- [3] D. Xiao *et al.*, “Burden of chronic obstructive pulmonary disease and its attributable risk factors in 204 countries and territories, 1990–2021: results from the Global Burden of Disease Study 2021,” *BMJ Open Respiratory Research*, 2025. [Online]. Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12815125/>
- [4] A. Agusti *et al.*, “The global burden of COPD: epidemiology and effect of prevention strategies,” *The Lancet Respiratory Medicine*, 2024. [Online]. Available: [https://www.thelancet.com/journals/lanres/article/PIIS2213-2600\(24\)00339-4/abstract](https://www.thelancet.com/journals/lanres/article/PIIS2213-2600(24)00339-4/abstract)
- [5] X. Li *et al.*, “Chronic obstructive pulmonary disease across three decades: trends, inequalities, and projections from the Global Burden of Disease Study 2021,” *Heliyon*, 2025. [Online]. Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11973060/>
- [6] World Health Organization, “Asthma,” WHO Fact Sheet, 2024. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/asthma>
- [7] W. Liao *et al.*, “Global, regional, and national burden of asthma and its attributable risk factors from 1990 to 2019: a systematic analysis for the Global Burden of Disease Study 2019,” *Respiratory Research*, vol. 24, p. 169, 2023. [Online]. Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10288698/>
- [8] Global Asthma Network, “The global asthma report 2022,” Global Asthma Network, 2022. [Online]. Available: <http://globalasthmareport.org>
- [9] Forum of International Respiratory Societies, “Respiratory diseases in the world: Realities of today — opportunities for tomorrow,” FIRS Report, 2017. [Online]. Available: <https://www.thoracic.org/about/global-public-health/firs/resources/firs-report-for-web.pdf>
- [10] M. Sarkar, R. Bhardwaj, I. Madabhavi, and J. Khatwal, “Lung sounds,” StatPearls, NCBI Bookshelf, 2023. [Online]. Available: <https://www.ncbi.nlm.nih.gov/books/NBK537253/>
- [11] D. Garau *et al.*, “Global trends in idiopathic pulmonary fibrosis mortality rates during 2001–2022,” *ERJ Open Research*, vol. 11, no. 6, 2025. [Online]. Available: <https://publications.ersnet.org/content/erjor/11/6/00362-2025>
- [12] National Heart, Lung, and Blood Institute, “Lung sounds,” NIH / NHLBI, 2023. [Online]. Available: <https://www.nhlbi.nih.gov/health/lung-sounds>

- [13] H. Melbye, T. O. Dybwad, T. S. Ingebrigtsen, N. Kanaan, and K. L. Fieten, “Prevalence and clinical associations of wheezes and crackles in the general population: the Tromsø study,” *BMC Pulmonary Medicine*, vol. 19, p. 173, 2019. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6739986/>
- [14] C. K. Pham *et al.*, “Respiratory sound classification for crackles, wheezes, and rhonchi in the clinical field using deep learning,” *Scientific Reports*, vol. 11, p. 17186, 2021. [Online]. Available: <https://www.nature.com/articles/s41598-021-96724-7>
- [15] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017. [Online]. Available: <https://arxiv.org/abs/1706.03762>
- [16] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, “An image is worth 16x16 words: Transformers for image recognition at scale,” in *International Conference on Learning Representations*, 2021. [Online]. Available: <https://arxiv.org/abs/2010.11929>
- [17] Y. Gong, Y.-A. Chung, and J. Glass, “AST: Audio spectrogram transformer,” in *Interspeech*, 2021, pp. 571–575.
- [18] L. D. Mang, F. D. González Martínez, D. Martinez Muñoz, S. García Galán, and R. Cortina, “Classification of adventitious sounds combining cochleogram and vision transformers,” *Sensors*, vol. 24, no. 2, p. 682, 2024.
- [19] N. Gavriely, Y. Palti, and G. Alroy, “Spectral characteristics of normal breath sounds,” *Journal of Applied Physiology*, vol. 50, no. 2, pp. 307–314, 1981. [Online]. Available: <https://doi.org/10.1152/jappl.1981.50.2.307>
- [20] Y.-L. Chen *et al.*, “The unreliability of crackles: insights from a breath sound study using physicians and artificial intelligence,” *npj Primary Care Respiratory Medicine*, vol. 34, 2024. [Online]. Available: <https://doi.org/10.1038/s41533-024-00392-9>
- [21] M. Sarkar, R. Bhardwaj, I. Madabhavi, and J. Khatwal, “Lung sounds,” StatPearls, NCBI Bookshelf, 2023. [Online]. Available: <https://www.ncbi.nlm.nih.gov/books/NBK537253/>
- [22] N. Jakovljevic, M. Tajber, A. Vidensky, and N. Stevenson, “Detection of respiratory phases in a breath sound and their subsequent utilization in a diagnosis,” *Applied Sciences*, vol. 11, no. 14, p. 6535, 2021. [Online]. Available: <https://doi.org/10.3390/app11146535>
- [23] B. M. Rocha, D. Filos, L. Mendes, I. Vogiatzis, E. Perantoni, E. Kaimakamis, I. Chouvarda, P. Carvalho, N. Maglaveras, and R. P. Paiva, “A respiratory sound database for the development of automated classification,” in *Precision Medicine Powered by pHealth and Connected Health*. Springer, 2018, pp. 33–37. [Online]. Available: https://doi.org/10.1007/978-981-10-7419-6_6
- [24] Y. Zhang *et al.*, “Advances and challenges in respiratory sound analysis: A technique review based on the icbhi2017 database,” *Electronics*, vol. 14, no. 14, p. 2794, 2025.
- [25] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, “An image is worth 16x16 words: Transformers for image recognition at scale,” in *International Conference on Learning Representations*, 2021. [Online]. Available: <https://arxiv.org/abs/2010.11929>

- [26] Y. Gong, Y.-A. Chung, and J. Glass, “AST: Audio spectrogram transformer,” in *Interspeech*, 2021, pp. 571–575.
- [27] P. Foret, A. Kleiner, H. Mobahi, and B. Neyshabur, “Sharpness-aware minimization for efficiently improving generalization,” in *International Conference on Learning Representations*, 2021. [Online]. Available: <https://arxiv.org/abs/2010.01412>